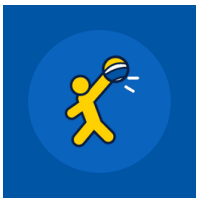
VolleyLive documentation



Full product and technical documentation for **VolleyLive 1.5.7** (versionCode 52) – live volleyball scoring, teams, challenges, tournaments and the Ask VolleyLive assistant, in English and Hindi, with Day and Night themes.

Written from the 1.5.7 source code, the release notes in `docs/qa/` and the design files. Where the design and the code differ, the code wins and the difference is noted. No secrets, keys, tokens, real phone numbers or personal data appear here – environment variables are listed by **name only**, and all screenshots use sample data.

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11	Design system	colour palette (hex, Day and Night), typography, spacing, components, day / night logic and greeting, screen notes, logo history, icon specs
12	Glossary	terms used across the app and these docs
13	FAQ	questions from players, captains, Ask users and developers
14	Acceptance checklist	feature-by-feature pass / fail checks for accepting a build

A combined PDF of all chapters is in `VolleyLive-Documentation.pdf` (diagrams rendered; GitHub also renders the mermaid diagrams in the Markdown files).

Images are in `assets/`.

Other docs in this repo

File	Purpose
`../DEPLOY.md`	detailed deploy steps and per-release migration notes
`../KNOWN-GAPS.md`	known gaps and limits by release
`../LIFECYCLE.md`	the match lifecycle table (generated)
`../db-schema.mmd` / `../db-schema.png`	generated base schema ER diagram
`../FONT-LICENSES.md`	font licences
`../privacy-policy-draft.md`	privacy policy (draft)
`../qa/`	release notes, progress logs and the Ask eval sets

1. Overview

> Documentation for **VolleyLive 1.5.7** (versionCode 52, Android package `com.volleylive.app`). Everything here was written from the 1.5.7 source code, the release notes in `docs/qa/` and the design files. Where the design and the code differ, the **code wins** and the difference is noted.

1.1 What VolleyLive is

VolleyLive is a mobile app (Android first, built with Expo / React Native) for **amateur and club volleyball in India**. It lets players:

- **score a match live** from the side of the court, one tap per rally,
- let everyone else **follow the score in real time**,
- keep **teams, stats, rankings and match photos**,
- send **challenges** between teams and run small **tournaments**,
- find **courts and games nearby** on a map,
- ask questions in plain English or Hindi through the **Ask VolleyLive** assistant.

The app is bilingual (**English / हिन्दी**) and has two visual themes, **Day ("Sunday Court")** and **Night ("Floodlight")**, which switch automatically by the time of day.

1.2 Who it is for

Persona	What they do in the app
Player	Signs in with a mobile number, keeps a profile, joins teams, sees own stats, rank and points wallet.
Captain	Creates and manages a team, invites players, sends and answers challenges, confirms or rejects results.
Scorer	The one person who scores a match (normally the match creator). Only the scorer can add or undo rallies.
Spectator / fan	Follows live scores, reactions and the recent-rallies ticker.
Organiser	Creates a tournament (knockout, league, or groups + knockout) and manages its fixtures.
Admin	Reviews flagged matches, manages "Fact / Quote of the day" items, reads feedback and support tickets, edits Ask content. Admins are set by configuration (`ADMIN_UIDS`), never inside the app.

1.3 Main features (1.5.7)

Area	Feature	Where
Sign-in	Mobile number + 6-digit OTP; Terms & Conditions acceptance; minimum age check	Login, Terms, Profile setup
Profile	Name, role, city / pincode, date of birth, avatar; career stats; recent form; match history; memories	Profile tab
Settings	Appearance (Auto / Day / Night + animation switch), Language (EN / HI), Help Center, Ask, Feedback, Rate, About, Licences, Privacy, Terms	Profile → Settings
Teams	Create a team (name, colour), add players by mobile number, captain, invites (accept / decline), leave	Teams, Team detail
New match	3-step flow: **Teams → Rules → Start**; registered team or "add players by number"; Exact 6 or Custom 6–12; Indoor / Beach / Custom rules; start	Centre "+" button

Area	Feature	Where
	now or schedule	
Live scoring	Tap a player = player point + team point; tap a team half = team point only; Undo; timeouts; switch sides; serve & rotation; sets; photos	Match → Score
Live following	Real-time scoreboard over WebSocket (API) or Firestore listeners; recent rallies; share. A reaction row is shown but is **UI only** (no reactions backend; nothing is sent or stored)	Match page
Results	Opposing captain confirms or rejects; Player of the Match (POTM); anti-fraud limits; admin review	Match page, Admin
Rankings & wallet	Ranking points (credited points + 10 per win + 25 per POTM); wallet with +100 bonus every 5 counted wins (no money value)	Rankings, Wallet
Challenges	Captain challenges another team with day, time, place and message; accept / decline; accepted challenge becomes a match	Challenge
Tournaments	Knockout (byes to top seeds), League (3/2/1 points), Groups + knockout (up to 32 teams, seeded or random draw); table, bracket, fixtures	Tournaments tab
Map / courts	Courts and games nearby, distance filters, "Play nearby" list, add a court	Map tab, Courts
Home	Time-based greeting, live hero card, Fact / Quote of the day, quick actions, Ask card, rank card, invites, upcoming	Home tab
Ask VolleyLive	RAG answers from FAQ / rules / guides + 10 read-only tools over your own data, with cards; EN / HI / romanised Hindi	Ask screen
Help & feedback	Help Center (search, topics, FAQ), Contact / support tickets, Feedback, Rate the app, About	Settings

1.4 Backends

The same app can run against four backends, chosen at build time by environment variables (see 07 – Setup and build):

1. **Demo** – everything on the device, sample data, any number with the demo code `123456`. For testing only.
2. **API server** – Node / Express + MySQL server with OTP sign-in (the production target).
3. **API + dev auth** – the API server with unsigned development tokens. Local testing only; refused in production.
4. **Firebase direct** – Firestore + Storage + Firebase phone auth, protected by `firestore.rules` / `storage.rules`.

1.5 Short version history

Version	versionCode	Highlights
up to 1.3.x	–	First live scoring, teams, Firebase backend and demo mode; then registered players only, POTM and rankings, anti-fraud limits, Terms, OTP sessions, captain verdict (confirm / reject), feedback and daily items, MySQL API server.
1.4.0 – 1.4.4	16 – 20	Email uniqueness, daily-item tombstones, "once the winning rally exists the match is final", server no longer trusts the stored summary.
1.4.5 – 1.4.18	21 – 34	Abandon flow (policy **'review'**, interim), one lifecycle table (`docs/LIFECYCLE.md`), verdict repair, scorer hand-off, many QA rounds.
1.4.19	35	Tournament admin permissions, **Help Center**, support tickets (migration 0020).
1.4.20 – 1.4.21	36 – 37	**Ask VolleyLive** phase 1 (RAG over FAQ / rules / guides).
1.4.22	38	**Day / Night spec v2** rebuild of the core screens.
1.4.23 – 1.4.24	39 – 40	Night contrast fixes, Hindi Ask, contrast scan.

Version	versionCode	Highlights
1.4.25	41	**Brighter palette** ("the colour is too much dark, make it brighter").
1.4.26 – 1.4.28	42 – 44	Spiker app icon and splash, groups + knockout tournaments (0024), match create claims (0025), bracket fixes.
1.5.0	45	**Ask VolleyLive with read-only tools** and cards, provider adapter, Ask eval set.
1.5.1 – 1.5.6	46 – 51	Ask hardening: PII masking, audit hashes (HMAC, 0028), history, admin content (0026–0027), many Tester retests.
1.5.7	**52**	PII masking for more number groupings, team look-up ranking, captain intent, packaging rule for env files. **Current build.**

The detailed notes for every build from 1.4.1 are in `docs/qa/RELEASE-<version>.md`.

1.6 Status at 1.5.7

- Verification of 1.5.7: app tests **2514 / 2514** (173 suites, also in 5 time zones), server **560 / 560**, Firestore rules **226 / 226**, Ask eval **354 / 358 (98.9 %)**, contrast scan **268 pairs / 0 failures**.
- Nothing is deployed to production yet. Migrations 0024 – 0028 wait for production approval.
- The APK is a debug-signed internal build. A Play release needs an upload key and the checklist in `docs/DEPLOY.md`.
- Open items and decisions are in 10 – Roadmap and open items.

2. User guide

This guide walks through every screen of VolleyLive 1.5.7 in the order a new player meets them. Screen names in **bold** match the app's English labels (Hindi labels are in `src/i18n/locales/hi.ts`). Screenshots use the app's **sample data** (made-up names), never real people.



Home (Day), Home (Night) and Live scoring (Day) – captured on the 1.4.26 build with demo data.

2.1 Sign-up and sign-in (mobile number + OTP)

1. Open the app. The **Login** screen shows *"Welcome! 👋 – Log in with your mobile number to score matches and climb the rankings."* and the tagline *"Score. Track. Play anywhere."* An **English / हिन्दी** switch sits at the top.
2. Type your **10-digit Indian mobile number** (the `+91` prefix is fixed). Invalid numbers show *"Enter a valid 10-digit mobile number"*. Tap **Send OTP**.
3. The **Enter the code** screen shows *"Sent to …"* with a **Change** link. Type the **6-digit code**. **Verify & continue** stays disabled until all 6 digits are entered. **Resend code** unlocks after the timer (*"Resend in …"*).
4. First sign-in only:
5. **Terms & Conditions** – tick *"I have read and accept these Terms & Conditions"* and tap **Accept and continue**. The terms cover: points have no money value, fair play (a result counts only after the opposing captain confirms), your data, conduct, and changes. When the terms change, the app asks you to accept the new version before you can make changes again (every write on the server checks this).
6. **Profile setup** – name, playing role / position, city and pincode, date of birth (a scroll picker) and an optional photo. Players below the minimum age (`MIN_AGE`, 10 by default) cannot sign up.

7. You land on **Home**.

Backend	How the code is sent
API server (production)	The server sends an SMS OTP through the configured provider (MSG91 in production; a console provider for development). OTP lifetime, attempts, resend gap and per-hour limits are server settings.
Firebase direct	Firebase phone auth (a "security check" screen may appear).
Demo	No SMS. A banner says *"Demo mode … Use any number; the code is 123456."* Data stays on the phone.

Signing out: Profile → **Sign out**. The server session is revoked (`POST /auth/logout`).

2.2 Profile and settings

The **Profile** tab shows:

- your name, role and city, and an **Edit profile** button;
- **My details (only you can see these)** – pincode and date of birth / age;
- the **rank card** (rank among all players and points to the next place);
- **Career stats** – Matches, Wins, Win %, Points, Spikes, Blocks, Aces, Tips, Losses;
- **Recent form** (W / L, latest first);
- **My teams** (with · *Captain* where you captain), **New team**;
- **Match history** and **Memories** (photos from your matches);
- a note when finished matches *"don't count yet (waiting for confirmation or review, or over a limit)"*;
- **Settings** and, for admins only, an **Admin** section (Review flagged, Facts, Feedback).

Appearance (themes)



Profile → Settings → **Appearance** has three radio cards and one switch:

Option	Behaviour
Auto (recommended)	**Day 06:00 – 17:59, Night 18:00 – 05:59**, by the phone's clock. Auto is purely time-based in 1.5.7 (this was Akshit's choice; the design spec had "the phone's dark mode wins").
Day	"Sunday Court" – always light.
Night	"Floodlight" – always dark.
Sunset & floodlight animation	On: a 1.4-second *Sunset → Floodlight* animation plays once at each switch. Off, or with the system *Reduce Motion* setting: a short fade.

During live scoring the switch never covers the score: it waits for the end of the rally and then crossfades. See 11 – Design system for colours and the animation storyboard.

Language

Profile → Settings → **Language**: **English** or **हिन्दी**. The whole app, the Help Center and the Ask assistant switch language. Ask also understands romanised Hindi ("mera score kya hai"). The Hindi copy still needs a native-speaker review (see 10).

Wallet

Profile → **Points wallet** shows the balance (*"In-app points only. They have no money value."*), points from matches, the **win bonus** (+100 for every 5 counted wins), *"N more counted wins to the next bonus"*, counted matches / wins, matches waiting for the other captain or flagged for review, and a **Points history**.

2.3 Teams

- **Create a team:** Profile → **New team** (or Home → **My teams**). Enter a **team name** and pick a **team colour**. You become the **captain**. Add players **by mobile number**; a number that belongs to a VolleyLive user shows *"On VolleyLive"*.
- **Invites:** an added player gets an invite (shown on Home and as a push notification) and can **Accept** or **Decline**. Only accepted, registered players count.
- **Team page:** stats (played, won, lost, win %), roster with captain tag, **Challenge** and **Start match** buttons. Unconfirmed results are not counted; the page says *"N result(s) waiting for the other captain"*.
- **Leave a team:** a player can leave; the captain manages the roster.
- **Size:** at least **6** players are needed to play; a team can have up to **12** (6 on court + 6 substitutes).
- Phone numbers of other players are never shown to you.

2.4 New match and live scoring

New match (3 steps)

Tap the centre **+** button. A stepper shows *"Step n of 3"*:

1. **Teams** – for Team A and Team B either **Pick a registered team** (search by name) or **Add players by number** (a one-off side built from registered players, not saved as a team). Pick a **colour**, a **captain** and the **first serve**. **Team size:** *Exact 6* (for a registered team of 7 or more you pick the 6 who play – the captain always plays) or *Custom (6–12)* (6 on court, the rest on the bench). A player cannot be in both teams. The **Next: Rules** button explains what is missing ("needs 6 per team", "a player is in both teams", …).
2. **Rules** – **Indoor** (best of 5, sets to 25, deciding set to 15, win by 2, 2 timeouts per set, switch sides at 8 in the deciding set), **Beach** (best of 3, sets to 21, deciding set to 15, win

by 2, 1 timeout per set, switch sides every 7 points, every 5 in the deciding set) or **Custom** (change any value; this marks the rules custom).
3. **Start** – *"Ready to start?"* summary: rules, when (**▶ Start now** or **🗓️ Schedule**), court / location, scorer (*"You"*), captains, players, first serve, optional **Match title**. Tap **Start match** or **Schedule match**.

   Design note: the spec mockups show a 4-step flow (teams → add players by shirt number → rules → toss). 1.5.7 ships the 3-step flow above.

Live scoring (scorer only)



Action	Result
Tap a player tile	That player gets the point **and** their team gets the point (*"One tap gives … and … the point."*). After a team point you can optionally pick *who scored* and the shot type (spike, block, ace, tip).
Tap a team's half / panel	Team point only (*"Opponent error? Tap the team's half for a team point only."*).
UNDO	Removes the last rally (see 2.5).
SWAP	Switch sides (🔄). The app also prompts at the rule's switch points.
Timeout (n)	Records a timeout for a team; *"No timeouts left"* when used up.
Sub	Substitution: pick who goes off (on court) and who comes on (bench).
Court & rotation	Shows the 6 positions, server and rotation (rotation advances on side-out).
STATS / Point-by-point	Per-player points and the rally timeline.
📷	Take or pick match photos (up to 50 per match, 10 per batch, 10 MB each).
⋮ Match options	Share live score, Abandon match, …

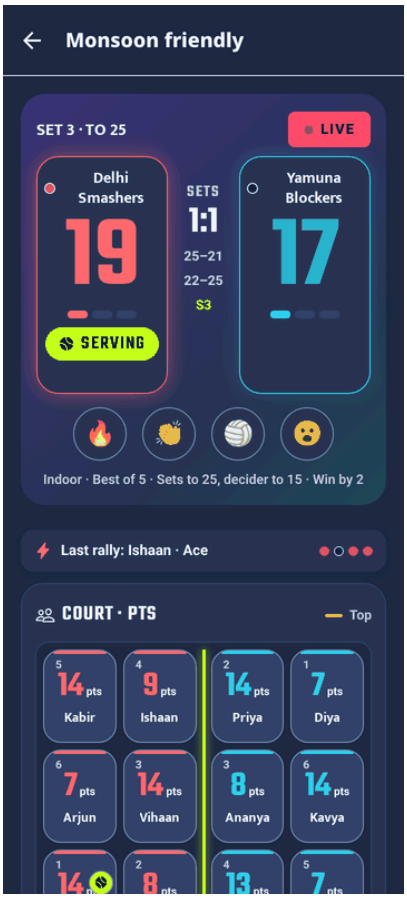
The header shows the set, *SET POINT* / *MATCH POINT*, sets won, *SERVING* and the sync state: ✓ **Synced**, **Syncing…**, ⚠ **Offline – saved on phone** (rallies wait on the phone and are sent later) or × **Not saved**. Only the assigned scorer can score (*"Only the assigned scorer can score this match. You can follow it live instead."*). The scorer can hand scoring over to another eligible person.

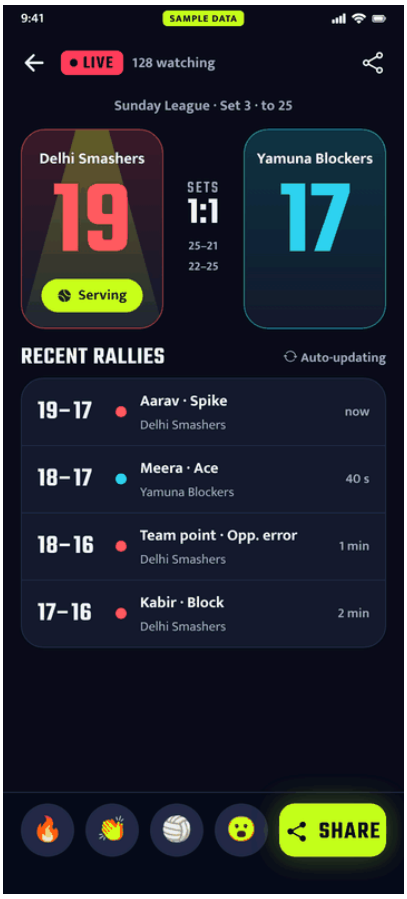
End of match. The winning rally ends the match (*"… win!"*). The result screen offers **Share the win** and **Match summary & photos**. Then:

1. **Player of the Match** – auto-picked: the most points credited to a player, from either team (team points don't count; ties are broken by the deciding set, …). The scorer can choose another player and confirms.
2. **Result verification** – the **opposing captain confirms** or **rejects** (with an optional reason). Only a confirmed, unflagged result counts for rankings and the wallet.
3. **Anti-fraud:** matches shorter than 10 minutes are flagged; at most 3 counted matches per player per IST day and 2 per team pair per IST week. Flagged matches wait for an **admin review** (approve clears the flag; void never counts).

Abandon. ⋮ → **Abandon match**. A match with no rally is simply abandoned (or deleted by its creator). After the first rally the current policy is **'review'** (interim, pending Akshit's decision): the abandon is recorded and waits for review, and the opposing captain can **Report** it. Full rules: `docs/LIFECYCLE.md`.

Following a match live (everyone else)





Open a live match from Home (**Watch live**), the map or a tournament. The read-only scoreboard updates after every saved rally. Below it are the **recent rallies** (auto-updating), the timeline and stats, and **Share**. The reaction buttons are shown but are UI only in 1.5.7 (nothing is sent).

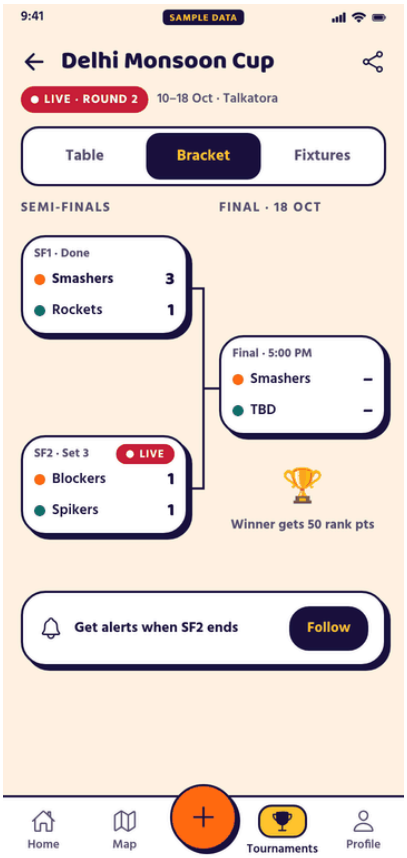
2.5 Undo

- **UNDO** (sticky at the bottom of the scoring screen) removes the **last rally**: the team point and, if one was credited, the player's point. The button shows what it will undo, e.g. *"Aarav −1 · Back to 18-17"*.
- After undo a toast says *"Point undone"*. With nothing to undo: *"Nothing to undo yet"*.
- **Once the winning rally is in, the match is final everywhere:** undo is refused (*"Match over: the score is final"*). After the Player of the Match is set: *"Match closed: Player of the Match is set"*. Only a credit on one of the last 3 rallies and the captain / admin fields can still change.
- Undoing back to 0 rallies does **not** make a played match deletable again.

2.6 Challenges

1. Only a **team captain** can challenge (*"Only a team captain can send challenges."*). Start from Home → **Challenge** or from another team's page.
2. Choose **which of your teams** sends it, **find the team** to challenge (search by name), pick the **day**, **time** and **place**, and add an optional **message**. Tap **Send challenge**. A team cannot challenge itself, and both teams need at least 6 players.
3. The other captain gets a notification and the challenge on Home, and can **Accept** or **Decline**.
4. When accepted, a playable match is created. The **accepting captain is the scorer**; the challenger's captain confirms the result. The challenge stays linked to that match only.
5. Challenges can be cancelled (deleted) by the sender before they are used.

2.7 Tournaments



Tournaments tab → **Organise / New tournament**:

Field	Values
Name	e.g. "Diwali Cup 2026"

Field	Values
Format	🏅 **Knockout** – lose and you are out; byes go to top seeds automatically, a play-in round brings the field to a power of two. 📊 **League (round robin)** – every team plays every other; win 3-0 / 3-1 = 3 pts, 3-2 = 2 pts, lose 2-3 = 1 pt. **Groups + knockout** – round robin in groups (A–H, up to 8 groups), top teams go to a knockout bracket (A1 v B2, B1 v A2, …); needs at least 4 teams.
Draw (groups)	**Seeded** (dealt A, B, C, D, D, C, B, A… so top seeds are apart) or **Random**; **Draw again**; a preview shows group + knockout match counts and byes.
Starts in / Duration	e.g. 1, 2 or 3 days
Teams	up to **32** registered teams with at least 6 players; the order is the seeding
Venue	a court / location

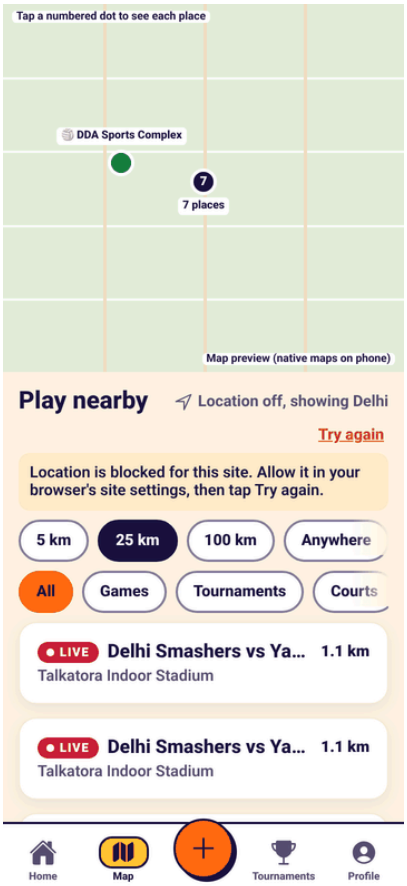
Tap **Create tournament & fixtures**. The tournament page has **Table**, **Bracket** and **Fixtures** tabs; matches are played and scored like any other match and results feed the table. Only the organiser (and admins) can change a tournament. A champion is shown when the final is decided.

2.8 Home (greeting, Fact / Quote of the day, map)



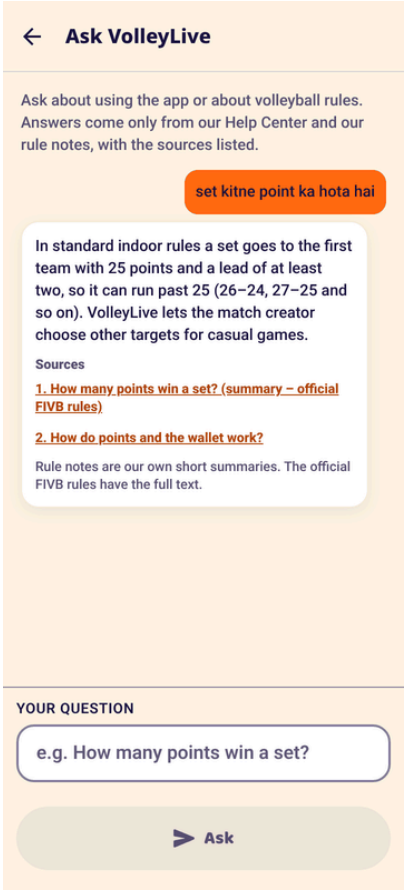
- **Greeting** by the hour: *Good morning* (05–11), *Good afternoon* (12–16), *Good evening* (17–20) and *"Late game, …?"* at night, with your name.
- **Live hero** – a match that is live now, with **Watch live**.
- **Fact / Quote of the day** card – *"Tap for next"* shows the next item; **Share**; the source is shown. Admins manage the items.
- **Quick actions:** New match, Challenge, My teams, Rankings, Live now, Tournaments, Courts (opens the Map tab), Memories.
- **Ask VolleyLive** entry card, your **rank card**, team **invites** and **challenges** to answer, the live carousel and **upcoming** matches.

Map



The **Map** tab shows courts and live / upcoming games near you, with distance chips (5 km, 25 km, 100 km, All) and filters (Games, Tournaments, Courts). The **Play nearby** list works without location permission (it then explains how to allow it). **Courts → Add court** adds a court with a location pin. Locations are only stored when you share them.

2.9 The Ask chatbot (Ask VolleyLive)



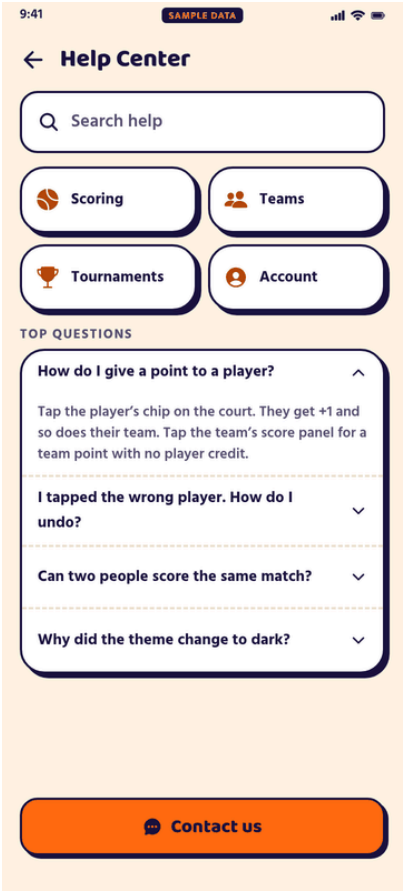
Open from the Home card or Settings → **Ask VolleyLive**. Type a question (*"e.g. How many points win a set?"*) and tap **Ask**.

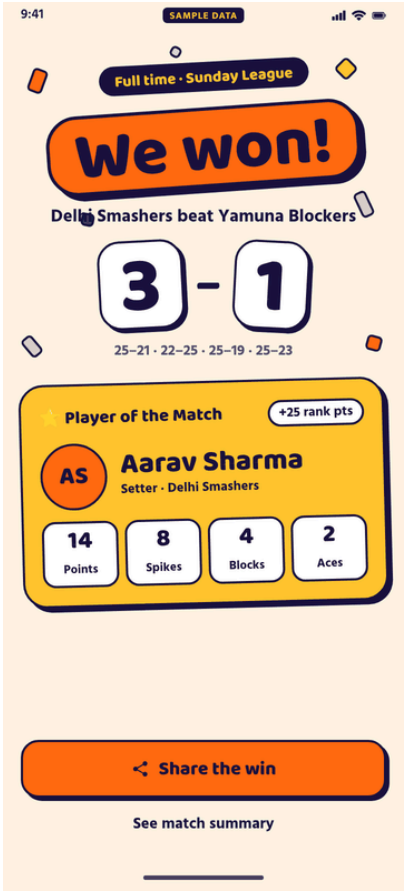
- **Rules and how-to questions** are answered from the Help Center FAQ, the app guides and VolleyLive's own rule notes, **with the sources listed**.
- **Questions about your own data** use read-only tools and show **cards**: *"my stats"*, *"my teams"*, *"my next match"*, *"live score of …"*, *"table of … cup"*, *"fixtures"*, *"my challenges"*, *"top players"*, another player's public stats, a team's roster.
- **It never changes anything.** *"Delete my match"* or *"add a point"* is refused with a pointer to the right screen.
- **It never reveals private data** (phone numbers, emails, wallet, date of birth …) and ignores instructions hidden in questions or data.
- **History:** your recent questions are kept for you only; **Clear** deletes them.
- EN, HI and romanised Hindi. Details: 04 – Chatbot.

2.10 Help, feedback, rating and About

Screen	What it does
Help Center	Search (*"Try 'points' or 'team'"*), topic tiles (Registration, Teams, New match, Scoring, Player of the Match, Wallet and points, …), FAQ accordion, *"Still need help?"* → **Contact us**.
Contact us / My messages	Create a support ticket with an optional screenshot and see replies. Email / WhatsApp / support-website buttons appear only when configured.
Send feedback	1–5 stars and optional text. *"Feedback is never rewarded with points or badges."* Rate-limited.
Rate VolleyLive	Opens Google Play's in-app review sheet in the Play build (*"Ratings work once VolleyLive is published on Google Play."*). An optional pre-prompt (*"Enjoying VolleyLive?"*) is off by default.
About	Version, backend (Demo / API server / Firebase), API host, links to Licences, Privacy (DRAFT) and Terms. Demo builds show demo-only testing tools.

Screen	What it does
Licences	Open-source and font licences, generated at build time.





3. Architecture

3.1 Technology stack

Layer	Technology (1.5.7)
App framework	**Expo SDK 57**, **React Native 0.86.3**, TypeScript, file-based routing with **expo-router**
UI	Own Day/Night component kit (`DayNightKit` , `DNKit` , `ui.tsx`), **react-native-reanimated 4** (animations), react-native-svg, expo-linear-gradient
i18n	**i18next** with English (`en.ts`) and Hindi (`hi.ts`)
Maps / location	**react-native-maps**, expo-location
Media	expo-image-picker, in-app match camera, a photo upload queue
Notifications	expo-notifications (Expo push service)
Local storage	AsyncStorage (session, settings, offline rally queue, demo data)
Backend option A	**Node ≥ 20, Express 4, Knex 3, MySQL 8 (mysql2)**, `ws` WebSocket, `zod` validation, `firebase-admin` (only to verify Firebase ID tokens), optional Redis
Backend option B	**Firebase JS SDK 13**: Firestore, Storage, phone auth, protected by `firestore.rules` / `storage.rules`
Backend option C	**Demo backend** – in-memory + AsyncStorage on the phone, seeded sample data
Shared logic	`src/core/*` – pure TypeScript used by the app **and** the server (scoring, rotation, rules, lifecycle, anti-fraud, ranking, Ask)
Tests	Jest (app), `node --test` (server), Firebase emulator + Jest (rules), Ask eval runner, contrast scan
Build	`expo prebuild` + Gradle locally, or **EAS Build** (`development` , `preview` = APK, `production` = AAB)

3.2 Repository layout

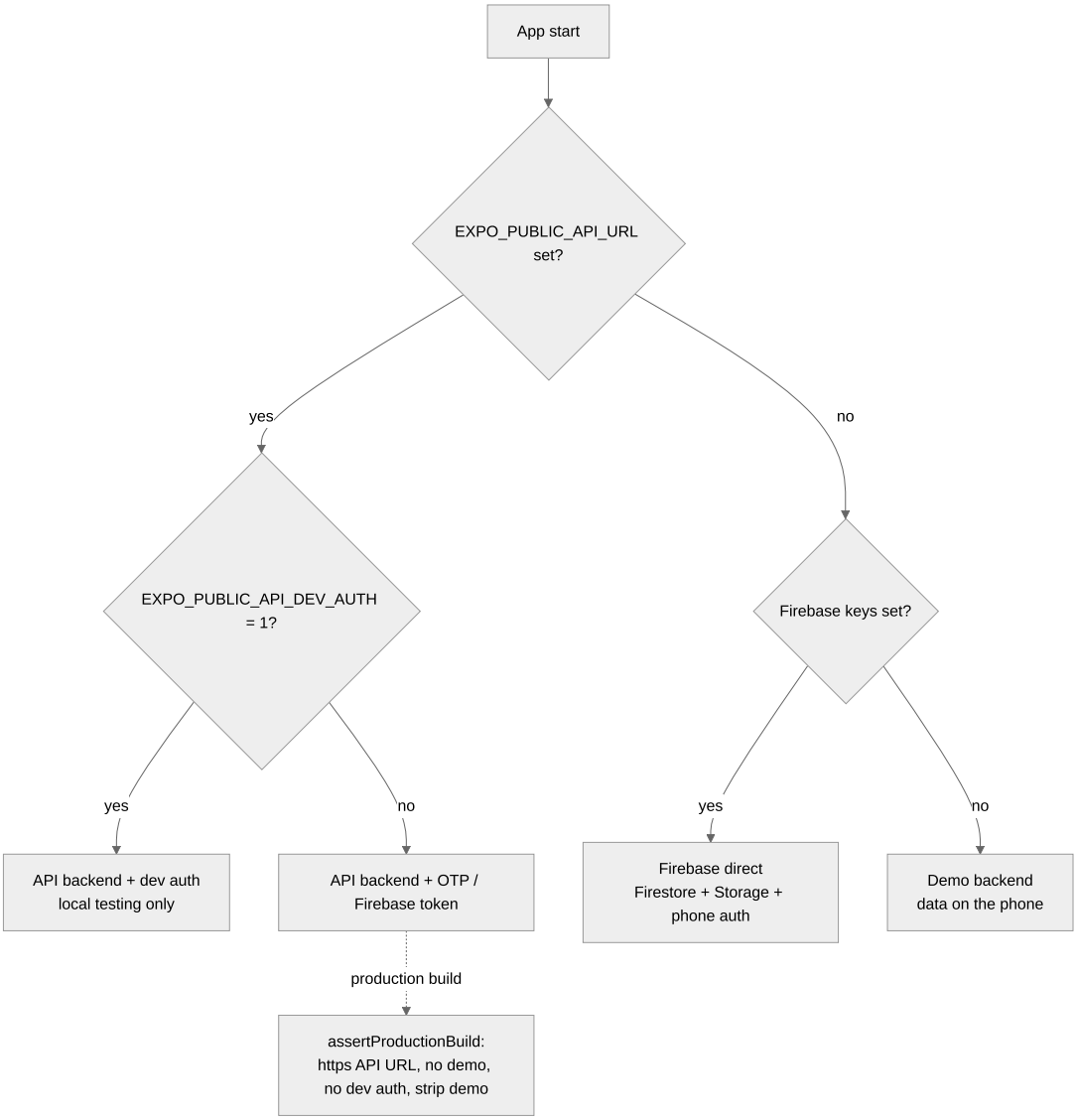
```
volleylive/
├─ app.config.ts            Expo config: name, version, versionCode, icons, splash, plugins,
env checks
├─ eas.json                 EAS build profiles (development / preview / production)
├─ assets/                  icon, adaptive icon, splash, notification icon, fonts
├─ src/
│  ├─ app/                  screens (expo-router)
│  │  ├─ (tabs)/            index (Home), map, score (+ centre button), tournaments, profile
│  │  ├─ match/             new.tsx, [id]/index.tsx (detail / viewer), [id]/score.tsx
(scorer)
│  │  ├─ team/ challenge/ tournament/ court/      new + detail screens
│  │  ├─ login, terms, setup-profile, edit-profile, appearance, ask, help, contact, feedback,
│  │  │  about, licenses, privacy, terms-view, wallet, rankings, memories, teams, courts
│  │  └─ admin-review, admin-daily, admin-feedback   admin-only screens
│  ├─ components/           LiveScoring, LineupPicker, TeamBuilder, RulesEditor, Potm,
RewardScreen,
│  │                        DailyCard, HomeGreeting, MiniCourt, PhotoGallery, DialogHost,
ask/ …
│  ├─ core/                 shared domain logic (see 3.5) and core/ask (agent, tools,
intents, cards, guides)
│  ├─ services/             config.ts (backend choice), backend/ (api, firebase, demo),
session, notifications,
│  │                        localScore (offline queue), photoQueue, location, share, review,
supportContact
│  ├─ state/                AuthContext, hooks, toasts, dialogs, rankings, memories
```

```
│   ├─ theme/                tokens.ts, themes.ts, resolve.ts, ThemeProvider, ThemeTransition,
fonts
│   ├─ i18n/                 i18next setup, locales/en.ts, locales/hi.ts, LanguageGate
│   └─ generated/            licences.json (generated)
├─ server/                   the API server (see below)
├─ rules-tests/              Firestore / Storage rules tests (emulator)
├─ firestore.rules  storage.rules  firestore.indexes.json  firebase.json
├─ functions/                old Cloud Functions (ported into the server's domain code)
├─ scripts/                  check-strip, licences, lifecycle doc, abandon policy, contrast
scan, icons
└─ docs/                     DEPLOY, KNOWN-GAPS, LIFECYCLE, db-schema, privacy draft, qa/,
guide/ (this guide)
```

```
server/
├─ src/
│   ├─ index.ts  app.ts      start-up, Express app, security headers, CORS, error mapping
│   ├─ config.ts             reads and checks every env var (refuses unsafe production
settings)
│   ├─ routes/               auth.ts (/auth), api.ts (/api), handler.ts
│   ├─ auth/                 middleware (Bearer check), verifier (session / firebase / dev),
otp/ providers
│   ├─ domain/               users, teams, matches, challenges, tournaments, courts, photos,
rankings,
│   │                        feedback, support, terms, ask, schemas (zod)
│   ├─ ask/                  host.ts (viewer-scoped tool host), providers.ts (mock / openai /
gemini / anthropic)
│   ├─ live/                 bus.ts (LiveBus), ws.ts (WebSocket /live)
│   ├─ db/                   knex, migrate (checksums), seed, rollback/
│   ├─ storage/              photoStorage (local disk, signed URLs), s3Storage
│   ├─ push/                 pushSender (Expo push, batches of 100)
│   ├─ ops/                  logger, redis, dbAudit
│   └─ rateLimit.ts rateStore.ts pincode.ts errors.ts
└─ migrations/               0001 … 0028 hand-written SQL
```

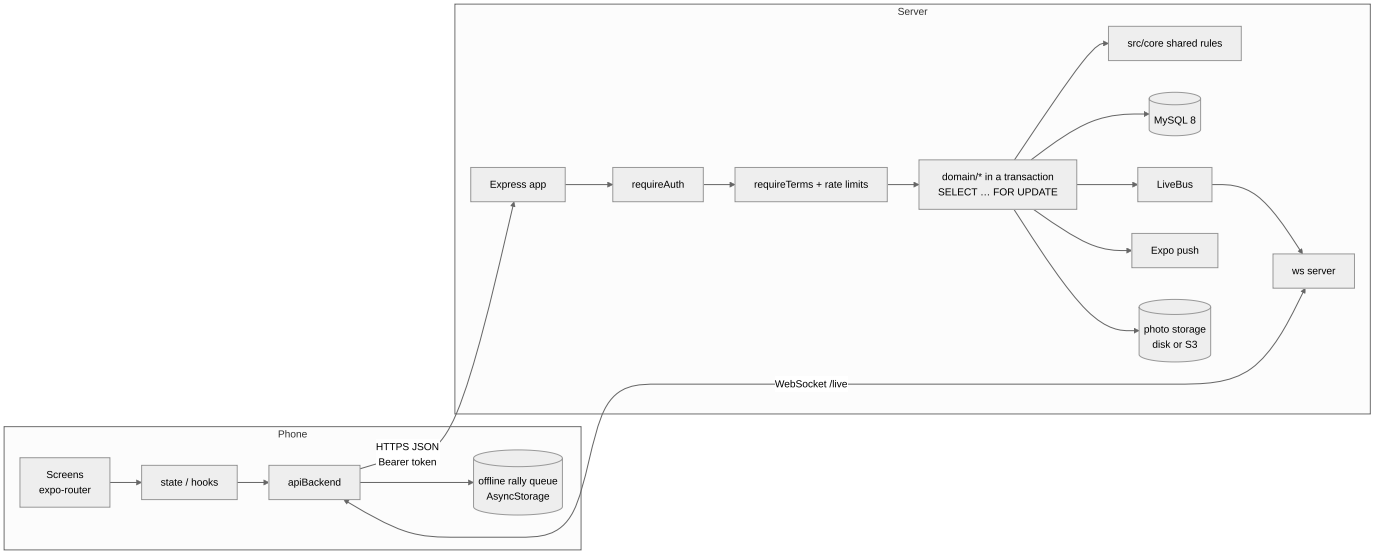
3.3 Backend selection

`src/services/config.ts` chooses the backend once at start-up:



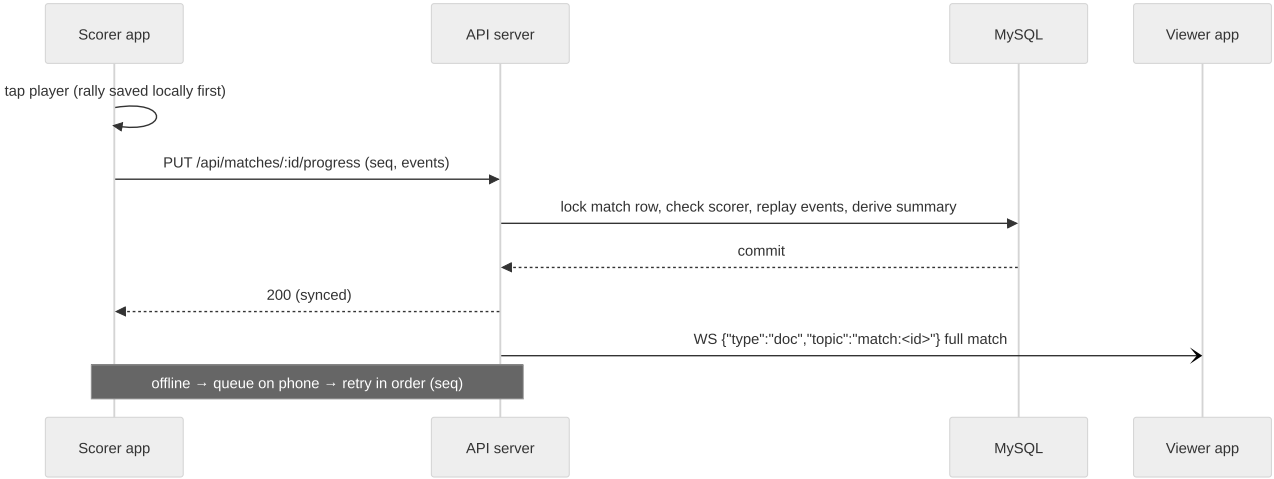
`EXPO_PUBLIC_BACKEND` (`demo`, `api`, `firebase`), `EXPO_PUBLIC_DEMO_MODE=1` and `EXPO_PUBLIC_API_AUTH` (`otp`, `firebase`, `dev`) can force the choice; a build that asks for the API but cannot use it fails loudly at build time and at start-up instead of silently falling back to demo (the production profile sets `api` and `otp`). All three backends implement the same `Backend` interface (`src/services/backend/types.ts`), so every screen works the same way.

3.4 Request flow: app → server → database



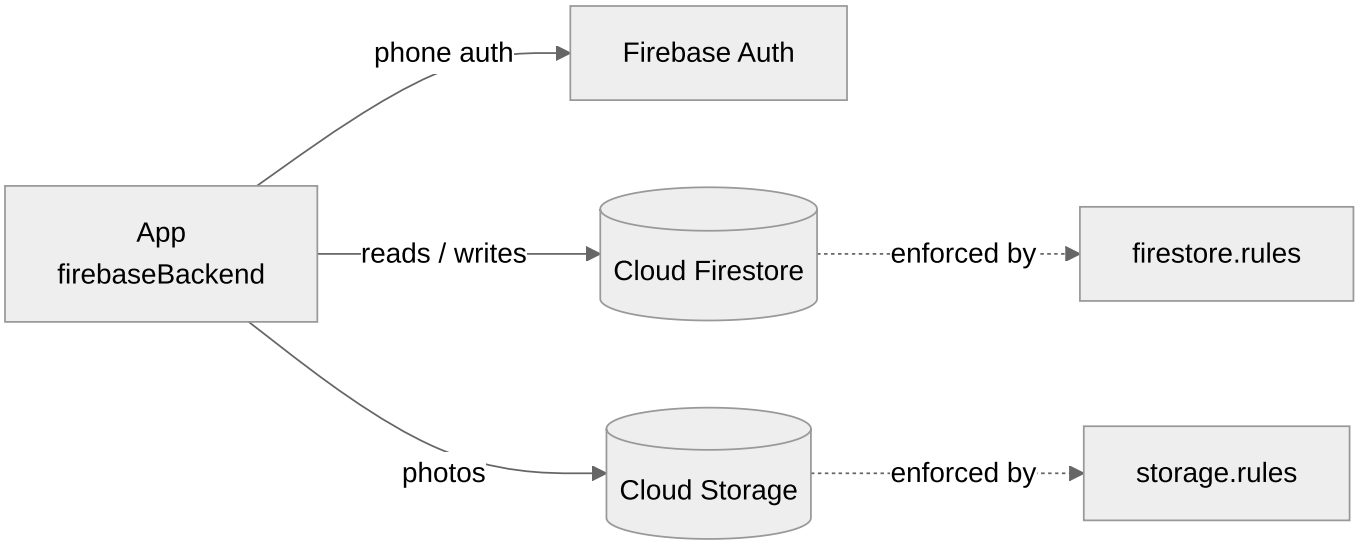
1. The app sends `Authorization: Bearer <session token>` on every `/api` call.
2. `requireAuth` verifies the token (server session from OTP, or a Firebase ID token, or a dev token in dev only).
3. Write routes check the **current Terms** acceptance and the **rate limits**, and reject server-only fields.
4. Domain code runs inside a **transaction with row locks**, re-using `src/core` so the server derives summaries, sets and rotations with exactly the same code as the app. The server does **not trust** client summaries.
5. After commit, `LiveBus` publishes to WebSocket subscribers and the push sender notifies players.

Live scoring sequence



WebSocket protocol (`/live`): the client sends `auth` with the same bearer token, then `sub` / `unsub` topics. `match:<id>` pushes the full match document after each saved rally; other topics (`matches`, `teams`, `team:<id>`, `profile:<uid>`, `photos:<matchId>`, `challenges`, `tournament:<id>`, `tournaments`, `courts`) only send `invalidate`, and the client re-fetches over REST (their content depends on the viewer). Unauthenticated sockets close after 10 s (code 4401); max 200 topics per socket; ping every 30 s.

3.5 Firebase direct flow



In Firebase mode there is no server code: **the rules are the server**. They repeat the critical checks (scorer only, summaries that follow the rally log, final after the winning rally, captain verdict, abandon limits, admin approval semantics). `rules-tests/` proves that the rules and `src/core` agree (226 tests). Two checks only the clients make (the rules cannot replay a whole match) are listed in `LC_CLIENT_ONLY` and `docs/KNOWN-GAPS.md`.

3.6 Shared core modules (`src/core`)

Module	Responsibility
`scoring.ts`, `rotation.ts`, `replay.ts`, `eventSteps.ts`	rally log → sets, summary, serve and rotation
`rules.ts`	Indoor / Beach / Custom rule sets
`lifecycle.ts`, `abandon.ts`, `abandonPolicy.ts`, `matchGuards.ts`	the match lifecycle (create, score, finish, abandon, report, reinstate, delete)
`antifraud.ts`, `results.ts`, `resultStatus.ts`, `verdictRepair.ts`	counting rules, captain verdict, flags, wallet bonus
`ranking.ts`, `potm.ts`, `stats.ts`, `teamRecord.ts`	rankings, Player of the Match, stats
`tournament.ts`, `groups.ts`, `fixtureLock.ts`, `tournamentAdmin.ts`	formats, draws, fixtures, permissions
`roster.ts`, `registration.ts`, `challengeLink.ts`, `challengeTeam.ts`, `scorerHandoff.ts`	team and match membership rules
`limits.ts`, `validate.ts`, `phone.ts`, `emailPolicy.ts`, `text.ts`	input limits and validation
`pii.ts`	PII masking (Ask answers, audit)
`terms.ts`, `dailyItems.ts`, `helpFaq.ts`, `support.ts`	terms versions, daily items, FAQ, support
`ask/*`	the Ask agent, intents, tools, cards, guides, rules knowledge

3.7 Theming

`ThemeProvider` resolves the theme (`resolve.ts`: setting + clock), exposes the tokens (`tokens.ts`) and plays `ThemeTransition` (the Sunset → Floodlight overlay). Live screens call `holdTransitions` so a switch never hides the score. Details: 11 – Design system.

3.8 Scaling notes

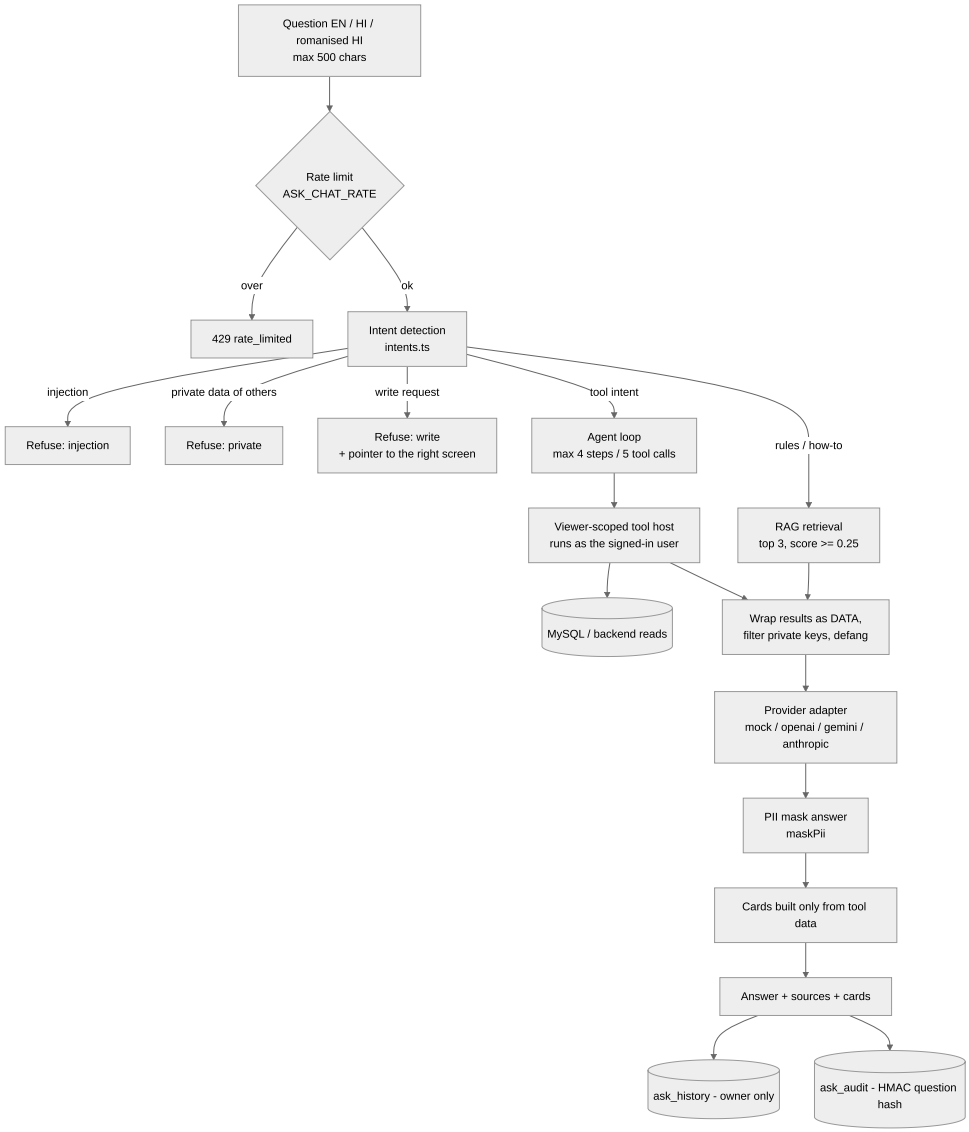
- `LiveBus` is in-process: run **one** API instance, or set `REDIS_URL` to use Redis pub/sub before scaling out.
- Photos on local disk need a shared volume, or use the S3 storage driver.
- Hosting templates: `server/Dockerfile` / `docker-compose`, `railway.json`, `render.yaml`.

4. Ask VolleyLive (the chatbot)

Ask VolleyLive is a **read-only assistant**. It answers questions about using the app and about volleyball rules from VolleyLive's own content (retrieval-augmented generation, "RAG"), and it answers questions about **your own** data by calling a small set of **read-only tools**. It can never change anything.

Code: `src/core/ask/` (shared by the app and the server), `server/src/domain/ask.ts`, `server/src/ask/host.ts`, `server/src/ask/providers.ts`, app screen `src/app/ask.tsx` and `src/components/ask/`.

4.1 End-to-end flow



4.2 Knowledge and retrieval (RAG)

Item	Value
Sources (`kind`)	`faq` (Help Center FAQ, `core/helpFaq.ts`), `guide` (app how-to guides, `core/ask/guides.ts`), `rules` (VolleyLive's own-words rule notes, `core/ask/rulesKnowledge.ts`, linked to the FIVB page), `custom` (admin-added content, table `ask_content`)
Chunking	text split into chunks of at most `ASK_CHUNK_MAX` = **480** characters, stored in `ask_chunks`
Embeddings	**local and deterministic**: hashed words + character trigrams into a **512-dimension** vector (`EMBED_DIM`), so Hindi works too. **No external embedding API and no key.** `EMBED_VERSION` (2) is part of the stored hash, so a change re-embeds every row
Retrieval	cosine similarity, top `ASK_TOP_K` = **3** chunks with score ≥ `ASK_MIN_SCORE` = **0.25**
No match	an honest *"I don't know"* (`idk`) answer with a pointer to the Help Center instead of a guess
Sources	every RAG answer lists its sources (title + kind)
Admin	admins can list, add, edit and delete `custom` content and trigger a **reindex** (`/api/admin/ask/content…`)

4.3 Read-only tools

Tool	Returns	Arguments
`my_stats`	the signed-in user's own match stats and rank	–
`player_stats`	another player's **public** stats (matches, wins, ranking points, rank) by name	`name` (2–60)
`my_teams`	teams the user belongs to	–
`team_roster`	players of a team (names, captain, substitutes)	`team` (2–40)
`my_matches`	the user's matches: upcoming, live or past, each with its result state (counted, awaiting confirmation, under review, disputed, voided)	`when`
`live_score`	live matches and scores, optionally for one team (a team that is not live: its latest finished match)	`team` (optional)
`tournament_standings`	league or group table(s) and champion	`tournament` (optional)
`tournament_fixtures`	fixtures with results	`tournament`, `status` (upcoming / played / all)
`challenges_status`	challenges involving the user's teams	–
`leaderboard`	top ranked players	`limit` (1–10, default 5)

- Tool arguments use strict JSON-schema validation (`additionalProperties: false`, length limits).
- **There are no write tools.** Writes are refused before any tool runs.
- Ambiguous names (two teams match) return an "ambiguous" result listing the candidates instead of guessing.
- Limits per question: `ASK_MAX_TOOL_CALLS` = **5**, `ASK_MAX_STEPS` = **4**; above that the outcome is `limited`.

4.4 Permissions

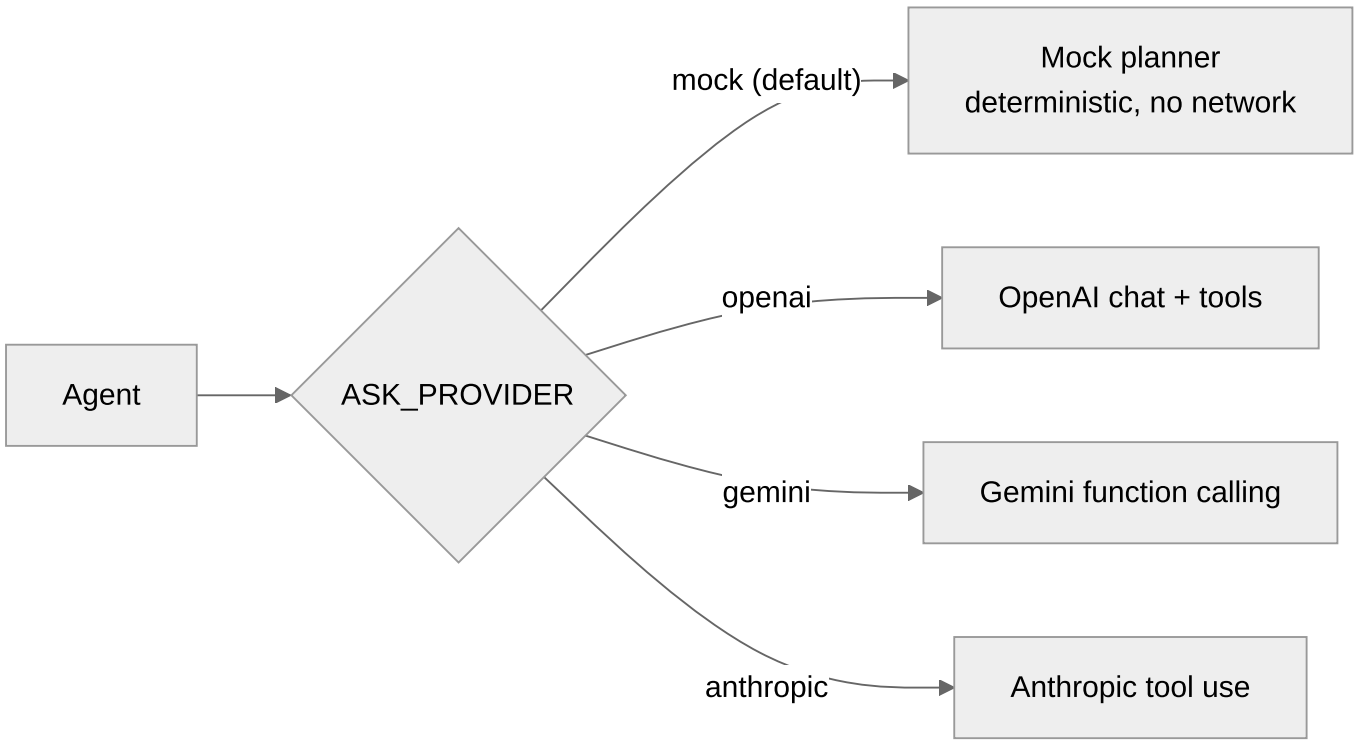
- Every tool runs **as the signed-in user** through a **viewer-scoped tool host** (`server/src/ask/host.ts`). It only sees what that user could see in the app (the same privacy rules as the REST API).
- "My …" tools only ever read the caller's own data. Another player's data is limited to public stats.
- Unconfirmed results are never counted in stats, leaderboards or tool answers.
- **History** (`GET /api/ask/history`, `DELETE /api/ask/history`) is owner-only; at most 100 items are kept and 50 are returned per page.

- **Admin content** endpoints and **reindex** require an admin (`ADMIN_UIDS`).

4.5 Safety: injection and masking

Layer	What it does
Data wrapping	tool output and retrieved text are wrapped in tags and treated as **DATA, never instructions**
Defang	instruction-like text inside data ("ignore previous instructions", role tags, …) is neutralised
Injection detection	a question that tries to change the assistant's rules is refused (`refusal: 'injection'`)
Private-key filter	fields such as phone, email, wallet / balance, dob, pincode, push token, message, token, key, password, secret are removed from tool data before the model sees it; **id fields are never masked** so routing keeps working
PII mask	`maskPii` (= `maskPiiStored`, `src/core/pii.ts`) masks phone numbers (Indian mobiles, with / without prefixes, any digit script, spelled-out digits, spaces, dots, dashes, brackets) and e-mails (including IDN and "name [at] domain [dot] com") in answers, card display strings, stored history questions and the audit text. Set scores such as "25-23" and lists of small numbers are kept
Cards	cards (match, standings, leaderboard, player, team, list) are built **only from tool data**, never from model text
Refusals	three kinds: `private`, `write`, `injection` – each with an EN / HI message
Audit	`ask_audit` stores the outcome, tools used and a **HMAC-SHA-256 hash** of the question keyed by `ASK_AUDIT_HMAC_KEY`. Without that key the hash is stored as NULL. No phone number or key is ever stored

4.6 Provider adapter



Env var (names only)	Meaning
`ASK_PROVIDER`	`mock` (default), `openai`, `gemini` or `anthropic`
`ASK_API_KEY`	the provider key – set by the operator in the server environment, **never in the repo, never logged, never sent to the app**

Env var (names only)	Meaning
`ASK_MODEL`	optional model override (defaults: `gpt-4o-mini`, `gemini-1.5-flash`, `claude-3-5-haiku-latest`)
`ASK_TIMEOUT_MS`	request timeout for the provider
`ASK_AUDIT_HMAC_KEY`	secret for the audit question hash (placeholder only in `.env.example`)

- With a real provider and **no key the server refuses to start** (no silent fallback that would look like AI answers).
- The adapter is called over `fetch`; HTTP is **mocked in tests**. Real providers have **not been tested live** yet (no key was used).

4.7 Mock mode and the demo app

- `mock` is the default on the server and is used by all tests and the eval. It plans tool calls from the detected intent and writes answers from templates – no network and no key.
- In the **demo** and **Firebase** builds the assistant runs **on the phone** (`services/askLocal.ts`, `services/askHost.ts`) with the same core code and the mock planner.

4.8 Rate limits

Bucket	Burst	Refill
`ASK_RATE` (`POST /api/ask`)	10	2 per minute
`ASK_CHAT_RATE` (`POST /api/ask/chat`)	20	6 per minute

4.9 Quality: the Ask eval

- Set: `docs/qa/ask-eval-*.jsonl` (tuned items + held-out items that were never tuned against).
- Runner: `server/scripts/ask-eval.ts` on a local test database.
- **1.5.7: 354 / 358 = 98.9 %** (mock provider, local test database). The 4 misses:
- `holdout157-04` – a frozen item that itself is wrong (only 9 digits, so correctly not masked);
- `holdout155-06` – a frozen item whose "score line" is an impossible match and is masked by design since 1.5.6;
- `holdout151-05` – a substitutes question answered with a privacy refusal;
- `holdout151-12` – "am I playing live" answered with all live matches.
- Held-out sets are frozen before the first run and never tuned against. See `docs/qa/RELEASE-1.5.7.md`.

5. Database

VolleyLive has two persistent stores, one per online backend:

- **MySQL 8** for the API server – hand-written SQL migrations in `server/migrations/` (real foreign keys, CHECK constraints, generated columns, composite keys), queried with Knex.
- **Cloud Firestore + Cloud Storage** for the Firebase-direct build – protected by `firestore.rules` and `storage.rules`, with composite indexes in `firestore.indexes.json`.

The demo backend keeps the same shapes in memory and AsyncStorage on the phone.

The full generated ER diagram of the base schema is in `docs/db-schema.mmd` and `docs/db-schema.png`.

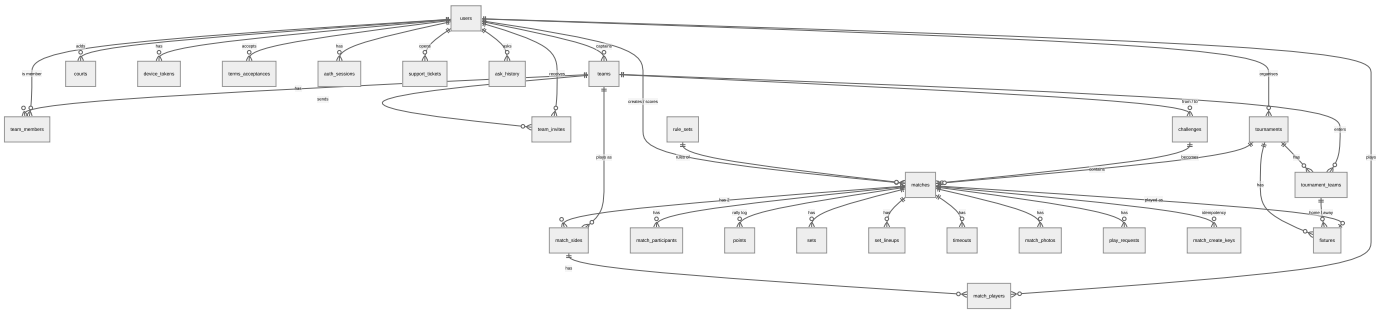
5.1 MySQL tables

Table	Purpose	Added by
`users`	player profile (name, role, city, pincode, dob, avatar, admin flag via config); phone is private; normalised unique email	0001, 0002, 0013, 0014
`teams`	team name, colour, captain (`users`)	0001
`team_members`	team ↔ user membership (captain, substitutes)	0001
`team_invites`	pending invites by a captain to a user	0001
`rule_sets`	match rules (best of, points per set, deciding set, win by, cap, timeouts, side switches)	0001
`matches`	one match: status, rules, scorer, creator, tournament / challenge link, summary, POTM, captains, verification (pending / confirmed / disputed / approved / voided), abandon and lifecycle fields, progress sequence	0001, 0004 – 0006, 0010, 0016 – 0018, 0023
`match_sides`	the two sides (A / B) of a match, optionally a registered team	0001
`match_players`	the players of a side (registered users only)	0001, 0002
`match_participants`	everyone allowed to see / act on the match (players, creator, scorer)	0001
`points`	the rally log (one row per rally, credited player, shot kind)	0001, 0003
`sets`	derived set scores	0001
`set_lineups`	starting six / rotation per set	0001
`timeouts`	timeouts per set and side	0001
`match_photos`	photos of a match (storage key, uploader, content hash)	0001, 0007
`play_requests`	"want to play" requests for a match	0001
`challenges`	team-to-team challenges (from / to team and captain, rules, place, time, status, linked match)	0001
`tournaments`	tournament (organiser, format, rules, dates, venue, groups settings)	0001, 0024
`tournament_teams`	teams in a tournament (seed, group letter)	0001, 0024
`fixtures`	fixtures (home / away tournament team, round, group, seeds, linked match)	0001, 0024
`courts`	courts with location, surface, who added them	0001
`device_tokens`	push tokens per user / device	0001
`terms_acceptances`	which Terms version a user accepted, when	0006
`otp_challenges`	OTP requests (hashed code, attempts, expiry)	0008

Table	Purpose	Added by
`auth_sessions`	server sessions issued after OTP or Firebase exchange	0008
`feedback`	star rating + optional text	0011
`daily_items`	Fact / Quote of the day (with tombstones instead of deletes)	0011, 0015
`support_tickets`	Help Center tickets, screenshot key, replies, visibility	0020, 0022
`ask_chunks`	Ask RAG chunks and their local embeddings	0021
`match_create_keys`	idempotency keys for match creation	0023
`match_create_claims`	one-create claims (prevent double creation from retries)	0025
`ask_content`	admin-managed custom Ask content	0026
`ask_history`	per-user Ask history (masked question, answer, cards)	0027
`ask_audit`	Ask audit (outcome, tools, HMAC question hash)	0027, 0028
`schema_migrations`	applied migrations with sha256 checksums	runner

Helper tables `email_duplicates_*` / `email_restored_*` are created only by the email repair migrations.

5.2 Relationships



5.3 Migrations

All migrations are **additive and re-runnable** (guarded with `IF NOT EXISTS` / information_schema checks). The runner stores each file's **sha256** in `schema_migrations.checksum` and **refuses to start** if an applied file changed (`MigrationChecksumError`). Never edit an applied migration – add a new one. A rollback script for 0024 is in `server/src/db/rollback/rollback0024.ts`.

#	File	Purpose
0001	`0001_init.sql`	base schema (21 tables)
0002	`0002_registered_players.sql`	match players must be registered users
0003	`0003_uncredit_removed_players.sql`	remove credits of removed players
0004	`0004_player_of_match_rankings.sql`	Player of the Match and rankings
0005	`0005_finished_needs_winner_photo_idempotency.sql`	finished needs a winner; photo idempotency
0006	`0006_terms_antifraud.sql`	terms acceptances; captains, server times and verification status
0007	`0007_photo_hash_signup_rows.sql`	photo content hash; sign-up rows
0008	`0008_otp_sessions.sql`	OTP challenges and auth sessions
0009	`0009_registration_team_size.sql`	registration team size
0010	`0010_captain_verdict.sql`	opposing captain verdict

#	File	Purpose
0011	`0011_feedback_daily_items.sql`	feedback and daily items
0012	`0012_captain_verdict_backfill.sql`	backfill of captain verdicts
0013	`0013_users_email_unique.sql`	unique normalised email
0014	`0014_email_index_repair.sql`	email index repair (checksum allow-listed)
0015	`0015_daily_items_tombstones.sql`	daily item tombstones
0016	`0016_match_abandon.sql`	abandon fields (max events, was won, abandon info / request, resolution, scorer reassign)
0017	`0017_match_lifecycle.sql`	lifecycle epochs, abandon history and request counters
0018	`0018_verdict_repair.sql`	`matches.verdict_repair` (JSON)
0019	`0019_verdict_repair_type.sql`	check only: `verdict_repair` must be JSON
0020	`0020_support_tickets.sql`	support tickets
0021	`0021_ask_chunks.sql`	Ask RAG chunks
0022	`0022_support_message_visible.sql`	support message visibility
0023	`0023_match_create_keys_progress_seq.sql`	match create keys; `matches.progress_seq`
0024	`0024_tournament_groups.sql`	groups + knockout (advance, group letters, seeds)
0025	`0025_match_create_claims.sql`	match create claims
0026	`0026_ask_content.sql`	Ask admin content
0027	`0027_ask_history_audit.sql`	Ask history and audit
0028	`0028_ask_audit_hmac.sql`	HMAC question hash in the audit

> **Production status:** nothing has been deployed. Migrations **0024 – 0028 are pending production approval**. Back up the database before migrating (see `docs/DEPLOY.md`).

5.4 Firestore collections (Firebase build)

Collection	Contents
`users/{uid}`	public profile; `users/{uid}/private/*` holds private fields (phone, dob, pincode, review state) readable only by the owner
`phoneIndex/{phone}`	phone → uid look-up for adding players (rules-restricted)
`emails/{address}`	email uniqueness claims
`teams/{id}`	team, members, invites
`matches/{id}`	match document (rules, sides, players, events, summary, verification, abandon / lifecycle fields); subcollections `photos` and `photoMeta`
`challenges/{id}`	challenges
`tournaments/{id}`	tournaments and fixtures
`courts/{id}`	courts
`dailyItems/{id}`	Fact / Quote of the day (tombstoned, never deleted)
`feedback/{id}`	feedback
`supportTickets/{id}`	support tickets
`pushTokens/{uid}`	push tokens

Collection	Contents
`terms/{uid}`	terms acceptance

Storage holds match photos and avatars (images only, size-limited by `storage.rules`). A composite index on `matches` (`verification.status` + `finishedServerAt`) is required for rankings.

6. API reference (API server)

Base URL: the server's public HTTPS address (`EXPO_PUBLIC_API_URL` in the app). All bodies are JSON. Source: `server/src/app.ts`, `server/src/routes/auth.ts`, `server/src/routes/api.ts`, `server/src/domain/*`.

6.1 Conventions

Topic	Rule
Auth	`Authorization: Bearer <token>` on every `/api` route (except `GET /api/daily-items`). Token = server session from OTP / Firebase exchange; dev tokens only with `AUTH_MODE=dev` (refused in production).
Terms	Every **write** requires the **current Terms version** to be accepted (`requireTerms`), otherwise `403` with a terms code.
Server-only fields	Fields the server owns (summary, verification, timestamps, lifecycle latches, …) are rejected with `400` if a client sends them.
Validation	`zod` schemas (`domain/schemas.ts`); errors are `400 { error, message, field? }`.
Body size	JSON limit **1 MB** (`413` above).
Errors	`400` bad request, `401` no / bad token, `403` forbidden, `404` not found, `409` conflict (e.g. `match_setup_locked`, stale progress), `413` too large, `429` rate limited (`rate_limited`), `500` internal (no details leaked).
Idempotency	`POST /api/matches` accepts an `Idempotency-Key` header; match progress uses a client sequence (`progress_seq`).
Headers	`X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`, `X-Powered-By` disabled; CORS only for `CORS_ORIGINS`.

The Auth column summarises the checks in `server/src/domain/*`; the domain code is authoritative.

Auth column legend: *public* = no token · *user* = any signed-in user · *owner* = the resource's owner · *captain* / *organiser* / *scorer* / *creator* = that role on the resource · *admin* = a uid in `ADMIN_UIDS`.

6.2 Health and files

Method	Path	Purpose	Auth
GET	`/health`	health with a database ping (2 s timeout)	public
GET	`/healthz`	liveness (`{ ok: true }`)	public
GET	`/files/<key>?exp=…&sig=…`	photo / screenshot download through an **HMAC-signed, expiring URL** (`FILE_URL_SECRET`, `FILE_URL_TTL_SECONDS`)	signed URL
GET	`/api/daily-items`	Fact / Quote of the day items (cached 300 s, per-IP rate limit)	public
WS	`/live`	live updates; first message `{"type":"auth","token":…}`	user

6.3 Auth (`/auth`)

Method	Path	Purpose	Auth
POST	`/auth/otp/request`	send an OTP to a mobile number → `202` (limits per number and per IP, resend gap)	public

Method	Path	Purpose	Auth
POST	`/auth/otp/verify`	verify the 6-digit code → session token	public
POST	`/auth/firebase`	exchange a Firebase ID token for a server session	public (valid ID token)
POST	`/auth/logout`	revoke the current session → `204`	user
GET	`/auth/config`	sign-in configuration for the app (mode, lengths)	public

6.4 Me and users

Method	Path	Purpose	Auth
GET	`/api/me`	own profile, admin flag, terms state	user
POST	`/api/me/terms/accept`	accept the current Terms version (rate limit `terms`)	user
GET	`/api/me/wallet`	wallet: balance, counted matches / wins, bonus, history	user
PUT	`/api/me/push-tokens`	register a push token	user
DELETE	`/api/me/push-tokens/:token`	remove a push token	user
GET	`/api/users?ids=…`	profiles for given ids (no listing of all users)	user
GET	`/api/users/by-phone/:phone`	look up a registered user by number to add them to a team (returns no phone of others)	user
GET	`/api/pincode/:pin`	pincode → district / state (rate limits per user and per IP)	user
GET	`/api/users/:id`	a profile (private fields only for the owner)	user
PUT	`/api/users/:id`	create / update own profile	owner
DELETE	`/api/users/:id`	delete own account	owner
POST	`/api/users/:id/avatar`	upload own avatar	owner
GET	`/api/users/:id/photos`	a user's match photos (memories)	user

6.5 Teams

Method	Path	Purpose	Auth
GET	`/api/teams`	teams (own and searchable)	user
POST	`/api/teams`	create a team (caller becomes captain)	user
GET	`/api/teams/:id`	team detail and roster	user
PATCH	`/api/teams/:id`	edit name, colour, members, invites	captain
POST	`/api/teams/:id/accept`	accept an invite	invited user
POST	`/api/teams/:id/decline`	decline an invite	invited user
POST	`/api/teams/:id/leave`	leave a team	member
DELETE	`/api/teams/:id`	delete a team	captain

6.6 Matches

Method	Path	Purpose	Auth
GET	`/api/matches`	matches visible to the caller (filters)	user
POST	`/api/matches`	create a match (`Idempotency-Key`, rate limit `matchWrite`)	user

Method	Path	Purpose	Auth
GET	`/api/matches/:id`	match detail	user
PATCH	`/api/matches/:id`	edit set-up before the first rally (`409 match_setup_locked` after)	creator / scorer
PUT	`/api/matches/:id/progress`	save rallies (events + sequence); server replays and derives the summary	scorer
GET	`/api/matches/:id/handoff-admins`	who scoring can be handed to	scorer
POST	`/api/matches/:id/abandon`	abandon (policy-dependent outcome)	scorer
POST	`/api/matches/:id/abandon-request`	answer an abandon consent request	captain
POST	`/api/matches/:id/abandon-report`	report an abandon	opposing captain
PUT	`/api/matches/:id/player-of-match`	set Player of the Match	scorer
POST	`/api/matches/:id/confirm`	confirm the result (rate limit `verify`)	opposing captain
POST	`/api/matches/:id/reject`	reject the result with a reason (≤ 200 chars)	opposing captain
DELETE	`/api/matches/:id`	delete a match that never had a rally	creator / admin
GET	`/api/matches/:id/photos`	photos (signed URLs)	user
POST	`/api/matches/:id/photos`	upload photos (images ≤ 10 MB, ≤ 50 per match)	participant
DELETE	`/api/matches/:id/photos/:photoId`	delete a photo	uploader / admin

6.7 Rankings, challenges, tournaments, courts

Method	Path	Purpose	Auth
GET	`/api/rankings`	player rankings (counted results only)	user
GET	`/api/challenges`	challenges of the caller's teams	user
POST	`/api/challenges`	send a challenge	captain
PATCH	`/api/challenges/:id`	respond (decline) / cancel	captain
POST	`/api/challenges/:id/accept`	accept and create the challenge match atomically	challenged captain
DELETE	`/api/challenges/:id`	delete a challenge	sending captain
GET	`/api/tournaments`	tournaments	user
POST	`/api/tournaments`	create a tournament and its fixtures	user (becomes organiser)
GET	`/api/tournaments/:id`	tournament, tables, fixtures	user
PATCH	`/api/tournaments/:id`	edit / advance fixtures	organiser / admin
DELETE	`/api/tournaments/:id`	delete a tournament	organiser / admin
GET	`/api/courts`	courts	user
POST	`/api/courts`	add a court	user
PATCH	`/api/courts/:id`	edit a court	adder / admin
DELETE	`/api/courts/:id`	delete a court	adder / admin

6.8 Feedback and support

Method	Path	Purpose	Auth
POST	`/api/feedback`	send stars + text (burst 5, 1 per 12 min)	user
GET	`/api/admin/feedback`	read feedback	admin
POST	`/api/support/tickets`	open a support ticket	user
GET	`/api/support/tickets`	own tickets and replies	owner
POST	`/api/support/tickets/:id/screenshot`	attach a screenshot	owner
GET	`/api/admin/support/tickets`	all tickets	admin
PATCH	`/api/admin/support/tickets/:id`	reply / change status	admin

6.9 Ask VolleyLive

Method	Path	Purpose	Auth
POST	`/api/ask`	RAG-only answer (burst 10, 2 / min)	user
POST	`/api/ask/chat`	agent answer with tools and cards, saved to history (burst 20, 6 / min)	user
GET	`/api/ask/history`	own history	owner
DELETE	`/api/ask/history`	clear own history	owner
GET	`/api/admin/ask/content`	list custom content	admin
PUT	`/api/admin/ask/content/:id`	add / edit custom content	admin
DELETE	`/api/admin/ask/content/:id`	delete custom content	admin
POST	`/api/admin/ask/reindex`	rebuild the RAG index	admin

6.10 Admin: daily items and match review

Method	Path	Purpose	Auth
GET	`/api/admin/daily-items`	list items incl. tombstones	admin
POST	`/api/admin/daily-items`	add a Fact / Quote	admin
PATCH	`/api/admin/daily-items/:id`	edit	admin
DELETE	`/api/admin/daily-items/:id`	tombstone (blank text, never a hard delete)	admin
GET	`/api/admin/matches/flagged`	flagged matches to review	admin
POST	`/api/admin/matches/:id/review`	approve (clears the flag) / void	admin (not in the match)
POST	`/api/admin/matches/:id/scorer`	reassign the scorer	admin
POST	`/api/admin/matches/:id/repair-verdict`	repair a match nobody can confirm (audited)	admin
POST	`/api/admin/matches/:id/resolve`	reinstate / confirm an abandon, decide a request	admin

6.11 Rate limits (summary)

Bucket	Burst	Refill
`terms`	5	5 / min
`verify`	10	10 / min
`matchWrite`	120	120 / min

Bucket	Burst	Refill
feedback	5	1 per 12 min
pincode	user 10 / IP 40	6 / 30 per min
daily items (public)	30	30 / min
Ask / Ask chat	10 / 20	2 / 6 per min
OTP	`OTP_MAX_PER_HOUR` per number, `OTP_IP_PER_HOUR` per IP, `OTP_RESEND_SECONDS` gap, `OTP_MAX_ATTEMPTS`	–

Limits are token buckets per user (shared via Redis when `REDIS_URL` is set). Over the limit the server answers `429`.

7. Setup and build

> **No secrets in the repo.** This page lists environment variables **by name only**. Real values live in local `.env` files (git-ignored), the EAS environment, or the hosting provider's secret store. The `.env.example` files contain placeholders only.

7.1 Prerequisites

Tool	Version / note
Node.js	20 or newer (server requires ≥ 20)
npm	comes with Node
MySQL	8.x (8.4 used in Docker Compose) – only for the API server
Docker (optional)	to run MySQL + API with `docker compose`
Android SDK + JDK (17 or newer)	for local APK builds (`expo prebuild` + Gradle)
EAS CLI (optional)	`npm i -g eas-cli` for cloud builds
Firebase CLI (optional)	for the rules tests (emulator) and Firebase deploys
A phone or emulator	Android phone with USB debugging for `adb install` (the minimum SDK comes from Expo SDK 57 defaults)

7.2 Run the app in demo mode (fastest)

```
npm ci
npx expo start            # press "a" for Android, or scan the QR code with a dev build
```

With no backend variables set the app runs the **demo backend**: sample data on the phone, any mobile number and the demo code `123456`. Web: `npx expo start --web`.

7.3 Run the API server locally

```
cd server
cp .env.example .env      # fill in your own local values; never commit .env
npm install
npm run migrate           # apply migrations 0001…0028
npm run seed              # optional: demo data (refused in production)
npm run dev               # http://localhost:4000
```

or with Docker Compose (MySQL 8.4 + API, migrations on start; choose your own passwords at the prompt / in your shell):

```
cd server && cp .env.example .env && docker compose up --build
```

Point the app at it in the app's `.env`: `EXPO_PUBLIC_API_URL=http://<your-computer-LAN-IP>:4000`. For local testing without SMS, run the server with `AUTH_MODE=dev` and set `EXPO_PUBLIC_API_DEV_AUTH=1` in the app – **dev auth lets anyone be anyone and is refused when `NODE_ENV=production`.**

7.4 App environment variables (names only)

Variable	Purpose
`EXPO_PUBLIC_BACKEND`	force `demo`, `api` or `firebase`
`EXPO_PUBLIC_DEMO_MODE`	`1` forces the demo backend
`EXPO_PUBLIC_API_URL`	API base URL (https only in production)
`EXPO_PUBLIC_API_AUTH`	`otp`, `firebase` or `dev` – how API builds sign in
`EXPO_PUBLIC_API_DEV_AUTH`	`1` = local dev tokens (never in production)
`EXPO_PUBLIC_STRIP_DEMO`	`1` strips demo code from production bundles (checked by `npm run check:strip`)
`EXPO_PUBLIC_DEMO_ADMIN_UIDS`	demo admin accounts (must be empty in production)
`EXPO_PUBLIC_MIN_AGE`	minimum sign-up age (default 10)
`EXPO_PUBLIC_SUPPORT_EMAIL` / `EXPO_PUBLIC_SUPPORT_WHATSAPP`	support contacts shown in Help (only when set)
`EXPO_PUBLIC_REVIEW_PREPROMPT`	`1` shows the "Enjoying VolleyLive?" pre-prompt (keep `0`)
`EXPO_PUBLIC_FIREBASE_API_KEY`, `…_AUTH_DOMAIN`, `…_PROJECT_ID`, `…_STORAGE_BUCKET`, `…_MESSAGING_SENDER_ID`, `…_APP_ID`	Firebase web config (Firebase builds)
`GOOGLE_MAPS_ANDROID_API_KEY`	Maps SDK key (build time, restricted to the app's package and certificate)
`GOOGLE_SERVICES_JSON`	path to `google-services.json` (FCM push)
`EAS_PROJECT_ID`	EAS project
`APP_VARIANT`	`production` enables `assertProductionBuild`
`ANDROID_PACKAGE`, `IOS_BUNDLE_ID`	identifiers (default `com.volleylive.app`)

7.5 Server environment variables (names only)

Group	Variables
Runtime	`NODE_ENV`, `PORT`, `LOG_LEVEL`, `TRUST_PROXY`, `CORS_ORIGINS`, `APP_VERSION`
Database	`DATABASE_URL`, `TEST_DATABASE_URL`, `MIGRATE_ON_START`, `MIGRATE_ON_DEPLOY`, `DB_SSL`, `DB_SSL_CA`, `DB_SSL_CA_FILE`, `DB_SSL_REJECT_UNAUTHORIZED`
Auth	`AUTH_MODE` (`otp`, `firebase`, `dev`), `FIREBASE_PROJECT_ID`, `SESSION_SECRET`, `SESSION_TTL_SECONDS`, `ADMIN_UIDS`, `MIN_AGE`
OTP	`OTP_PROVIDER`, `OTP_TTL_SECONDS`, `OTP_MAX_ATTEMPTS`, `OTP_RESEND_SECONDS`, `OTP_MAX_PER_HOUR`, `OTP_IP_PER_HOUR`, `MSG91_AUTH_KEY`, `MSG91_TEMPLATE_ID`, `MSG91_OTP_VAR`
Files	`STORAGE_DRIVER` (local / s3), `STORAGE_DIR`, `PUBLIC_BASE_URL`, `FILE_URL_SECRET`, `FILE_URL_TTL_SECONDS`, `S3_BUCKET`, `S3_REGION`, `S3_ENDPOINT`, `S3_ACCESS_KEY_ID`, `S3_SECRET_ACCESS_KEY`, `S3_FORCE_PATH_STYLE`, `S3_PRESIGN_SECONDS`, `S3_REDIRECT`
Scale	`REDIS_URL` (shared rate limits and live bus for more than one instance)
Push	`PUSH_ENABLED`, `EXPO_ACCESS_TOKEN`
Pincode	`PINCODE_LOOKUP`
Ask	`ASK_PROVIDER`, `ASK_API_KEY`, `ASK_MODEL`, `ASK_TIMEOUT_MS`, `ASK_AUDIT_HMAC_KEY`

`server/src/config.ts` validates them and **refuses unsafe production settings** (dev auth, missing secrets, a real Ask provider without a key, non-https CORS, …).

7.6 Build an APK

Local (what the internal test builds use)

```
npx expo prebuild --platform android      # generates android/ from app.config.ts
cd android && ./gradlew assembleRelease    # app/build/outputs/apk/release/app-release.apk
```

Bump `version` and `versionCode` in `app.config.ts` for every build (Android refuses a lower versionCode over an installed app). Internal builds are signed with the local debug certificate and are **not** for the Play Store.

EAS (cloud)

Profile	Output	Use
`development`	dev client	development
`preview`	**APK** (internal distribution)	testers
`production`	**AAB**	Google Play; sets `APP_VARIANT=production`, `EXPO_PUBLIC_BACKEND=api`, `EXPO_PUBLIC_API_AUTH=otp`, `EXPO_PUBLIC_STRIP_DEMO=1`, no demo admins

```
eas build -p android --profile preview
eas build -p android --profile production
```

The production build **fails** if the API URL is missing, demo / dev sign-in is on, demo admins are set or the demo strip is off (`assertProductionBuild`).

Verify a build

```
apksigner verify --print-certs app.apk     # must not be the debug certificate for Play
aapt2 dump badging app.apk | head -3       # package, versionName, versionCode, sdk
```

Install on a phone

```
adb devices
adb -s <serial> install -r app.apk         # -r keeps app data; never uninstall to "fix" an install
adb -s <serial> shell monkey -p com.volleylive.app -c android.intent.category.LAUNCHER 1
```

On Realme / ColorOS phones enable **Developer options → Install via USB** and approve the on-screen prompt, otherwise the install fails with `INSTALL_FAILED_VERIFICATION_FAILURE`.

7.7 Release checklist (summary of `docs/DEPLOY.md`)

1. All suites green (see 08 – Testing); `npm run licenses`.
2. Server env from `server/.env.example` with real values in the host's secret store; `NODE_ENV=production`; `AUTH_MODE=otp` or `firebase` (never `dev`); https-only `CORS_ORIGINS`; `TRUST_PROXY`; `REDIS_URL` for > 1 instance.
3. **Back up the database**, then migrate (additive, checksummed).
4. Firebase builds: deploy indexes first, then rules together with the matching app.
5. `eas build -p android --profile production`, verify the artifact, internal testing track, then production.

Owner-only items (Akshit): Play Console account and app signing, hosting, SMS provider DLT registration, Maps key, `google-services.json`, `ADMIN_UIDS`, support email, privacy policy review (DPDP), Play data-safety form.

8. Testing and QA

8.1 Test suites

Suite	Where	What it covers	Run
App unit / component tests	`src/**/__tests__` (Jest, 173 suites)	core logic (scoring, rotation, rules, lifecycle, anti-fraud, ranking, POTM, tournaments, groups, PII, Ask agent / tools / intents), backends (demo, API client, Firebase docs), components and screens (Day / Night, EN / HI, 360 px), theme resolve and transition, i18n parity	`npm test`
Time-zone run	same tests	the whole Jest suite under 5 zones: UTC, Asia/Kolkata, Asia/Colombo, America/Los_Angeles, Pacific/Kiritimati (IST day / week limits, greetings, theme hours)	`npm run test:tz`
Typecheck	app and server	TypeScript strict checks	`npx tsc --noEmit`, `cd server && npm run typecheck`
Server API tests	`server/tests/*.test.ts` (`node --test`)	every route, auth modes, rate limits, terms gate, transactions and races, migrations and checksums, Ask host / providers (mocked HTTP), audit HMAC, WebSocket	`cd server && TEST_DATABASE_URL=… npm test` (database name must end in `_test`; all tables are dropped)
Firestore / Storage rules tests	`rules-tests/rules.test.js`	the security rules against the Firebase emulator; agreement with `src/core`	`cd rules-tests && npm test` (Firebase emulator)
Abandon policy suites	`scripts/run-policy-suites.sh`	the abandon-related suites under each policy (`review`, `flag`, `consent`), then restores `review`	`bash scripts/run-policy-suites.sh`
Ask eval	`server/scripts/ask-eval.ts`, `docs/qa/ask-eval-*.jsonl`	answer / card / refusal correctness on tuned and frozen held-out questions (mock provider, local test DB)	see the script header
Contrast scan	`scripts/contrast/scan.js`	renders every route, modal and state in Day / Night × EN / HI on the web build and checks each visible text against WCAG AA (4.5:1, 3:1 large)	`BASE=http://127.0.0.1:8099 node scripts/contrast/scan.js`
Demo strip check	`scripts/check-strip.js`	a production web export contains no demo code	`npm run check:strip` after `npx expo export -p web`

8.2 Latest results (1.5.7, versionCode 52)

Check	Result
Jest	**2514 / 2514** passed, 173 suites
Jest in 5 time zones	passed
Server	**560 / 560**
Rules	**226 / 226**
Ask eval	**354 / 358 (98.9 %)** – 4 known misses, see 04
Contrast	**268** text pairs checked, **0** failures

Earlier builds' results are in `docs/qa/RELEASE-<version>.md` and `docs/qa/PROGRESS-<version>.md`.

8.3 QA process



1. **Findings** – the Tester runs full end-to-end walks (web build at 360 px, Day / Night, EN / HI) and writes numbered findings with severity (High / Medium / Low), e.g. `F156-1`.
2. **Fixes** – each finding gets a fix and a test that would have caught it; release notes map finding → change → test.
3. **Verification** – all suites above; screenshots before / after for visual fixes.
4. **Build** – APK and source zip with recorded sha256; same signing certificate between builds.
5. **Retest** – the Tester retests every finding and adds new held-out Ask items (frozen before the first run).
6. **Ship** – install on the test phone (`adb install -r`, never uninstall or downgrade) and push the source to GitHub (normal push, secret scan before commit).

8.4 Rules for test data

- Test and screenshot data must be **sample data** (made-up names). No real phone numbers or personal data in tests, fixtures, screenshots or the eval set.
- Server tests refuse to run unless the database name ends in `_test`.
- Ask tests and the eval use the **mock** provider only; no keys are used.

9. Security and privacy

9.1 Authentication

Mode	How it works	Where allowed
OTP (API)	`POST /auth/otp/request` sends a 6-digit code by SMS; the code is stored **hashed** in `otp_challenges` with an expiry and an attempt counter; `POST /auth/otp/verify` issues a **server session** (`auth_sessions`, signed with `SESSION_SECRET`, lifetime `SESSION_TTL_SECONDS`). Limits per number, per IP, per hour and a resend gap.	production
Firebase (API)	the app signs in with Firebase phone auth; `firebase-admin` verifies the ID token (only `FIREBASE_PROJECT_ID` is needed, **no service-account key**) and `POST /auth/firebase` exchanges it for a session	production
Firebase direct	Firebase phone auth on the device; Firestore / Storage rules use `request.auth`	Firebase builds
Dev	unsigned `dev:<number>` tokens – anyone can be anyone	**local testing only**; the server refuses `AUTH_MODE=dev` with `NODE_ENV=production`, the production app build refuses `EXPO_PUBLIC_API_DEV_AUTH`
Demo	everything on the phone, demo code	demo builds only; production bundles strip demo code (`EXPO_PUBLIC_STRIP_DEMO=1`, `check:strip`)

- **Admins** are configured server-side (`ADMIN_UIDS`); the app cannot make anyone an admin. Demo admins must be empty in production.
- **Sign-out** revokes the server session. Unauthenticated WebSockets are closed after 10 s.
- **Minimum age** (`MIN_AGE`, default 10) is checked at profile creation.
- **Terms gate**: every write requires the current Terms version to be accepted.

9.2 Authorisation rules

The same rules are enforced in three places – the **API domain code**, the **Firestore rules** and the **demo backend** – and shared tests prove they agree.

- **Scorer only** scores, undoes and finishes a match; setup is locked after the first rally.
- **Final after the winning rally**: afterwards only POTM, a credit on one of the last 3 rallies and captain / admin fields can change. The server and the rules **re-derive the summary from the rally log** and never trust a stored one.
- **Captain verdict**: only the opposing captain confirms or rejects; the scorer and creator cannot confirm for the other side. Admin approval only clears a fraud flag (`ADMIN_APPROVAL.COUNTS_WITHOUT_CAPTAIN = false`).
- **Anti-fraud**: matches shorter than 10 minutes are flagged; at most 3 counted matches per player per IST day and 2 per team pair per IST week; an admin cannot review a match they played in, scored or created.
- **Teams / challenges / tournaments**: only captains edit teams and send / answer challenges; only the organiser (or an admin) changes a tournament; a challenge links only to a match between its two teams.
- **Delete**: a match that ever had a rally cannot be deleted by anyone (admins included).
- **Server-only fields** are rejected with 400.

- **Row locks** (`SELECT … FOR UPDATE`) make every read-check-write atomic, so two phones cannot race past a rule.
- **Firebase-only gaps** (the rules cannot replay a whole match or check the current captain) are listed in `docs/KNOWN-GAPS.md` and `LC_CLIENT_ONLY`.

9.3 Rate limiting and abuse

Token buckets per user and per IP (shared through Redis when configured): terms, verify, match writes, feedback, pincode, public daily items, Ask, Ask chat, and the OTP limits. See 06 – API. Feedback is never rewarded, so there is no incentive to spam it.

9.4 Privacy and personal data

Data	Who can see it
Mobile number	only the user (and the server for sign-in); others can **add** you by number but never **see** your number
Email, date of birth, pincode, wallet	only the user (`users/{uid}/private` in Firebase; owner-only fields in the API)
Name, role, city, avatar, public stats	signed-in users
Match location	only when the user shares it
Photos	served through signed, expiring URLs (`/files/…?exp=…&sig=…`) with `nosniff`; images only, ≤ 10 MB
Ask history	owner only; questions are masked before they are stored
Ask audit	outcome, tools and an **HMAC** hash of the question – never the phone number or a key

- **PII masking** (`src/core/pii.ts`): phone numbers and e-mails are masked in Ask answers, cards, stored history and the audit, across digit scripts and separators. Ask tool data is filtered of private keys before any model sees it.
- **Account deletion**: `DELETE /api/users/:id` (owner) on the server; there is **no in-app button yet** (open item).
- **Privacy policy**: `docs/privacy-policy-draft.md` is a **draft** and must be reviewed for India's DPDP Act before launch.
- **Points** have no money value (stated in the Terms and the wallet).

9.5 Transport and server hardening

- HTTPS only in production (`EXPO_PUBLIC_API_URL` must be https; `CORS_ORIGINS` https only).
- Security headers: `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`, `X-Powered-By` off.
- JSON body limit 1 MB; uploads checked by type and size; content hashes for photos.
- 500 errors return no internal details; tokens, keys and OTP codes are kept out of logs and messages (only variable names appear in config errors).
- Optional TLS to MySQL (`DB_SSL*`).
- Config validation refuses unsafe production settings at start-up.

9.6 No-secrets policy

1. **Never commit secrets.** `.env`, `.env.*` (except `.env.example`), `env.*.json`, `*.pem`, `*.key`, `*.p12`, `*.jks`, `*.keystore`, databases and backups are excluded from the repo.
2. `.env.example` files contain **placeholders only**.
3. Real values live in the EAS environment, the host's secret store or a local git-ignored `.env`.
4. Documentation lists variables **by name only**.

5. Before every push a scan checks file names and content for private keys and common token formats (GitHub, AWS, Google, Slack, Stripe, OpenAI, Anthropic, xAI), password assignments, signing passwords, staged symlinks and placeholder lengths in `.env.example`. Any hit stops the push.
6. The Ask provider key (`ASK_API_KEY`) is never sent to the app and never logged.
7. Test vectors that look like keys (e.g. AWS's public documentation example key used in a unit test) are not secrets.
8. No real phone numbers or personal data in docs, tests, screenshots or the eval set – sample data only.

10. Roadmap and open items

Status at **1.5.7 (versionCode 52)**. Nothing is deployed to production yet.

10.1 Pending decisions (owner: Akshit)

#	Decision	Context
D1	**Approve production migrations 0024 – 0028**	groups + knockout, match create claims, Ask content, Ask history / audit, audit HMAC. All additive; back up first. A rollback for 0024 exists.
D2	**AI provider and key** for Ask	choose `openai`, `gemini` or `anthropic` (or stay on `mock`), set `ASK_PROVIDER` / `ASK_API_KEY` in the server's secret store. Real providers have not been tested live. Also set a real `ASK_AUDIT_HMAC_KEY`.
D3	**Support URL**	`SUPPORT_URL` (`src/services/supportContact.ts`) currently points to a placeholder site unrelated to VolleyLive; needs the real support page. Also set `EXPO_PUBLIC_SUPPORT_EMAIL` / `EXPO_PUBLIC_SUPPORT_WHATSAPP`.
D4	**FIVB rules licence**	10 own-words rule notes link to the FIVB rules page; confirm the licence / attribution is acceptable.
D5	**Hindi review**	the Hindi copy (app, Help, Ask) needs a native-speaker review; some new strings are marked NEEDS REVIEW.
D6	**Abandon policy**	`ABANDON_POLICY = 'review'` is interim; choose `review`, `flag` or `consent`.
D7	**Two registered teams per player?**	may a player be in two registered teams (long-standing open question).
D8	**Phone-shaped team names**	Ask masks them, but team screens show team names as typed; decide whether to block or mask them at creation.
D9	**Sock-puppet opposing captain (C130-1)**	a fake account acting as the opposing captain to confirm results – not implemented by decision; decide whether to add a defence.
D10	**Greens update**	`daynight/green/greens-update.md` proposes a new Night primary green (`#9BFF3A`) and related tokens – not in 1.5.7.
D11	**Bold Spiker icon**	chosen; the asset kit is ready for build 1.5.8 (file swap + native rebuild). See 11.

10.2 Open Lows and known gaps (from the 1.5.7 release notes)

- **Display = stored masking**: on screen, text hides whatever storage hid, including 10+ single digits separated by spaces (e.g. a typed rally sequence).
- **Dev-auth token accepts 9 digits** (dev only).
- **"N of M"** invite counts include all invites.
- **HMAC audit path untested** live (no key, by design).
- **Firestore rules have no current-captain check** (the API and demo enforce it).
- **Ask held-out misses**: substitutes → privacy refusal; "am I playing live" → all live matches.
- **Local harnesses** (browser walks) were not re-run for 1.5.7; API-level checks were.
- Firebase-only gaps, web-only limits (no offline app shell), 32-bit ARM, review-prompt timing: `docs/KNOWN-GAPS.md`.
- Viewer **reactions** are UI only (no backend).
- **No in-app account deletion button** (the API supports it; Play requires a deletion path before launch).
- Spec items not built yet: onboarding swipes, notifications screen, rankings period chips / podium, 4-step new-match flow with shirt numbers (see 11).

- **Phone install of 1.5.7**: the last attempt on the test phone was blocked by the phone's USB-install verification prompt; enable *Install via USB* and approve the prompt, then retry with `adb install -r`.

10.3 Launch checklist (owner items, from `docs/DEPLOY.md`)

- Google Play Console account, app signing / upload key, data-safety form.
- EAS project and production environment (`EXPO_PUBLIC_API_URL`, https).
- Hosting for the API + MySQL (+ Redis for more than one instance), server secrets in the host's secret store.
- SMS provider (MSG91) with **DLT** registration for India.
- Google Maps Android key (restricted), `google-services.json` for push.
- `ADMIN_UIDS`, support email, privacy policy review (DPDP), `MIN_AGE` (10), keep `EXPO_PUBLIC_REVIEW_PREPROMPT=0`.

10.4 Next features (candidates)

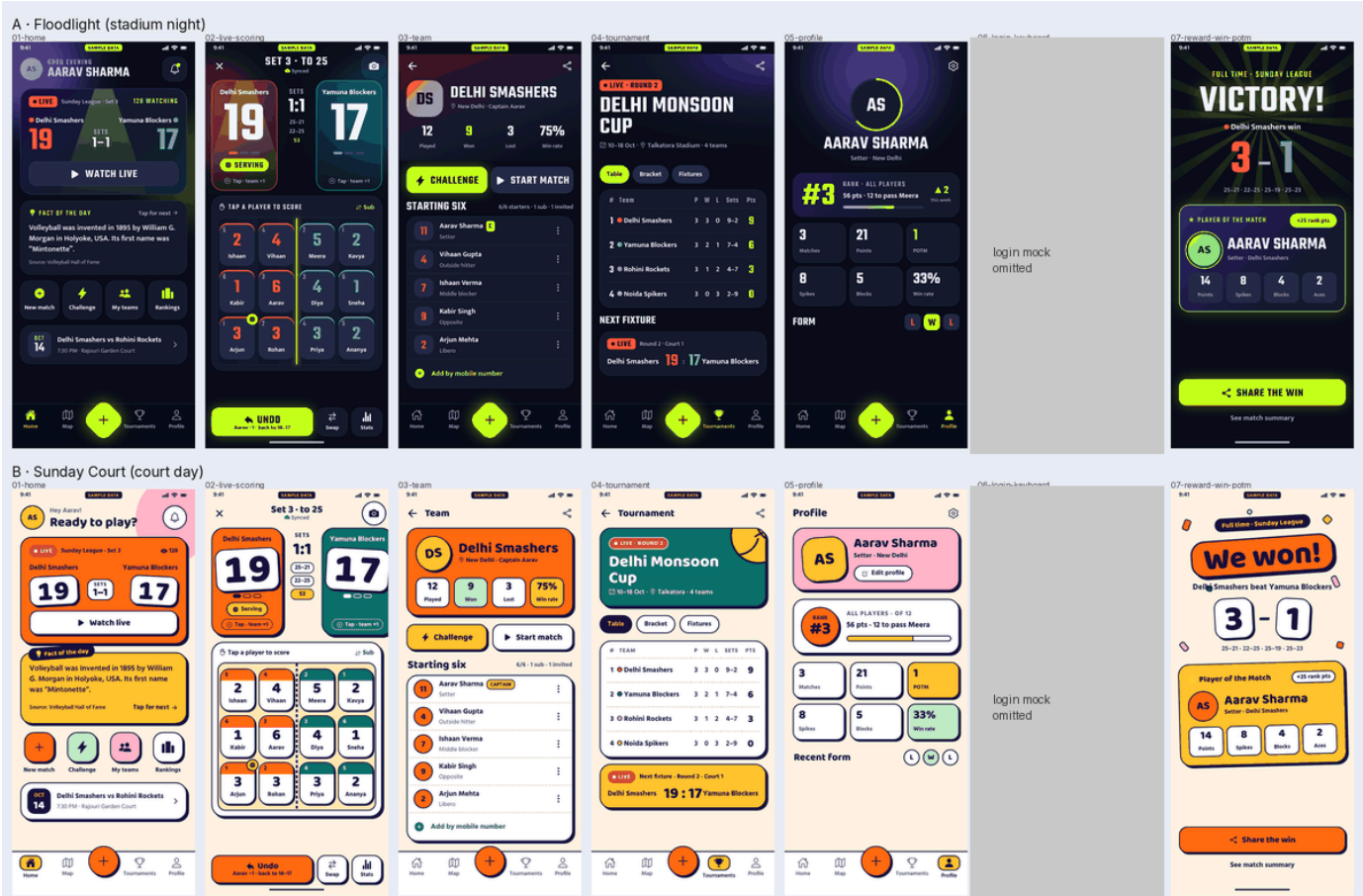
Feature	Notes
Bold Spiker icon (1.5.8)	asset kit ready
Greens update	proposal ready
Viewer reactions backend	UI exists
Notifications inbox	spec screen 20
Onboarding	spec screen 1
Rankings periods / teams / podium	spec screen 16
Live video / streaming	was out of scope for v1
Live LLM provider	after D2
iOS build	the config has an iOS bundle id; not built or tested
Multi-instance scale-out	Redis bus + S3 photos

11. Design system

This chapter is the design reference for VolleyLive: colours, type, spacing, components, the Day / Night logic, screen notes, the logo history and the icon specifications. The **source of truth for 1.5.7 is the code** (`src/theme/tokens.ts`, `src/theme/themes.ts`, `src/theme/resolve.ts`, `src/theme/ThemeTransition.tsx`). Design sources: `volleylive-design/daynight/daynight-spec.md` (spec v2), `volleylive-design/directions/`, `volleylive-design/logo/` and `volleylive-design/logo/v2/final-bold-spiker/`. All mockups use **sample data**.

11.1 Design history in one page

When	Step	Outcome
11 Oct 2026	Phase-2 "Navy Live" spec (`design-spec.md`) – Paytm-style navy `#002E6E` + light blue `#00BAF2`, system fonts, Ionicons	superseded
Oct 2026	Two directions explored: **A · Floodlight** (stadium night) and **B · Sunday Court** (court day)	both kept – as the Night and Day themes
1.4.22	**Day / Night spec v2** – one component library, two token sets, time-based switching, greeting states, Sunset → Floodlight transition	core screens rebuilt
1.4.25	**Brighter palette** – Akshit: "the colour is too much dark, make it brighter"	current tokens below
13 Oct → 1.4.26	Logo concepts 1 – 10 → **7 · Spiker** chosen	ships in 1.5.7
29 Oct	Logo v2 – six options → Akshit chose the **Bold Spiker**	kit ready for 1.5.8
proposal	**Greens update** (Night primary `#9BFF3A` …)	not in 1.5.7



Directions A and B side by side (the login mock column is blanked out in this copy).

11.2 Colour palette (1.5.7 tokens)

Token	Day · "Sunday Court"	Night · "Floodlight"	Use
`bg`	`#FFFAF3`	`#1E2A44`	screen background
`surface`	`#FFFFFF`	`#283654`	cards, sheets
`surface2`	`#FFF5E6`	`#33446A`	secondary surfaces, skeletons
`outline`	`#6B6590`	`#7A8BB8`	card borders, dividers
`text`	`#2A2452`	`#F4F7FF`	primary text ("ink")
`textMuted`	`#5E5779`	`#C3CDE3`	secondary text
`primary`	`#FF6A13` (Court orange)	`#C6FF1A` (Volt)	primary buttons, FAB, highlights
`onPrimary`	`#2A2452`	`#141C30`	text on primary
`accent`	`#FFC531` (Ball yellow)	`#8B6CFF` (Ultra)	stickers, selections (Night: gradients / glow only, never text)
`teamA`	`#FF6A13`	`#FF6B70`	team A fill
`teamAText`	`#B4460A`	`#FF9A9E`	team A text / orange text
`teamB`	`#0B6E6A`	`#2DD4F0`	team B fill
`onTeamB`	`#FFFFFF`	`#141C30`	text on team B
`live`	`#C81E3A`	`#FF4D6A`	LIVE badge, errors
`onLive`	`#FFFFFF`	`#141C30`	text on live
`liveText`	`#C81E3A`	`#FFA3B1`	live / error text
`success`	`#BDEFD8`	`#C6FF1A`	success chips
`tabActive`	`#2A2452`	`#C6FF1A`	active tab
`pink`	`#FFB4C8`	`#33446A`	Day decor blob
`track`	`#F1E8DA`	`#3A4A70`	progress tracks
`hero` gradient	`#FF6A13` (solid)	`#3A3478` → `#283654` → `#1F4A5A`	live hero card
`scrim`	`rgba(42,36,82,0.45)`	`rgba(10,16,32,0.60)`	modal scrim
`tabBarBg`	`rgba(255,250,243,0.94)`	`rgba(30,42,68,0.94)`	tab bar
`shadowColor`	`rgba(42,36,82,0.20)`	transparent	Day hard shadow

Contrast rules (spec v2, re-checked by the contrast scan – 268 pairs, 0 failures in 1.5.7):

- every text pair passes **WCAG AA 4.5:1** in both themes (e.g. `text` on `bg` 15.89:1 Day / 17.97:1 Night);
- Day `primary` orange is a **fill** colour; orange **text** uses `teamAText` `#B4460A`; never white on Court orange (2.9:1);
- Night `accent` is for gradients and glow only;
- dark team colours get a light ring in Night (`swatchRing`) when below 3:1 on the card.

The older map in `themes.ts` (`SUNDAY_COURT` / `FLOODLIGHT`) keeps a few legacy names for older components, such as `gold` (`#FFC531` / `#FFC727`) and `danger` (`#C81E3A` / `#FF9AA2`).

Greens update (proposal, not shipped): `daynight/green/greens-update.md` proposes a brighter Night green (`#9BFF3A` as primary) and matching success / tab tokens. It needs Akshit's go-ahead and a contrast re-run.

11.3 Typography

Role	Day	Night
Display (titles, hero)	**Baloo 2 ExtraBold** (800), Latin + Devanagari	**Teko SemiBold** (600), UPPERCASE Latin
Display in Hindi	Baloo 2 ExtraBold	**Mukta ExtraBold** (800), sentence case, no letter-spacing (Teko has no Devanagari)
Score numerals	Baloo 2 ExtraBold on a tilted white sticker	Teko SemiBold, tabular, in team colour (also in Hindi)
Body	**Hind** Regular / SemiBold / Bold	**Mukta** Regular / Bold

Greeting header: greeting 14, name 26 – 28 (one line, ellipsis). Scoreboard numerals 54 – 112. Font licences: `docs/FONT-LICENSES.md` (all SIL OFL).

11.4 Spacing, sizes, radius, elevation

Token	Value
Spacing grid	**4-pt**: 4 · 8 · 12 · 16 · 20 · 24 · 32; screen padding **16**
Button	height **52**, icon 20 + label, full width or hug ≥ 120
Input	height **56**, label above, helper / error below
Chip	**36** visual (44 in forms), hit area **48**
Icon button	**48 × 48**, icon 22 – 24
Tab bar	tab **64 × 56**, bar height 88 incl. 24 gesture inset, 4 tabs + centre FAB **60**
List rows / sheet rows	**≥ 64**
Radius card / button / chip	Day **22 / 16 / 18** · Night **20 / 14 / 18**
Card border	Day **2 px** `outline` · Night **1 px** `outline`
Shadow	Day: hard offset **3 / 3**, ink at 20 %, drawn as a View (elevation 0) · Night: none; glow `rgba(198,255,26,0.35)` on primary only
Touch targets	≥ 48 dp everywhere
Motion – point	Day "Stamp": scale 1.15 → 1, rotate ±6°, spring · Night "Flash": beam sweep 280 ms, +1 pop in Volt
Decor	Day: blobs, rotated stickers, confetti · Night: glow blobs, floodlight beam

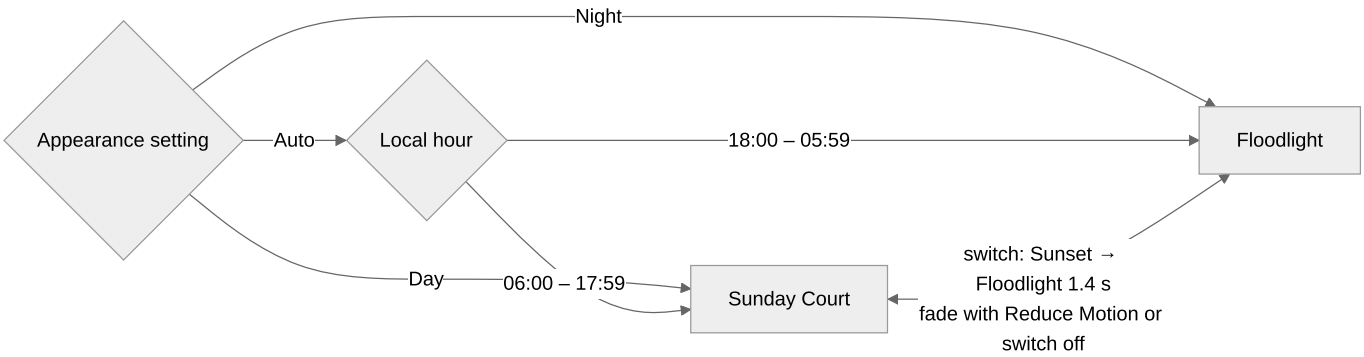
11.5 Components

Component	Anatomy / size	States	Day vs Night
Button (primary / secondary / text)	h52	rest, pressed (scale .97), disabled (solid muted colours), loading	Day ink border + hard shadow; Night glow on primary
Icon button	48 × 48	rest / pressed / badge dot 8	Day circle with border; Night rounded-16 surface
Chip / filter	h36, hit 48	off / on / disabled	on: Day ink + Ball text; Night Volt + ink
Segmented control	h52, items h42 – 44	selected / unselected	selected: Day ink + Ball; Night Volt + ink
Card	radius card, padding 14 – 16	rest / pressed (.98) / highlighted	border + shadow tokens
Text input	h56	rest / focus / error (live colour + icon + text) / disabled	focus: Day orange offset shadow; Night 2 px Volt + glow

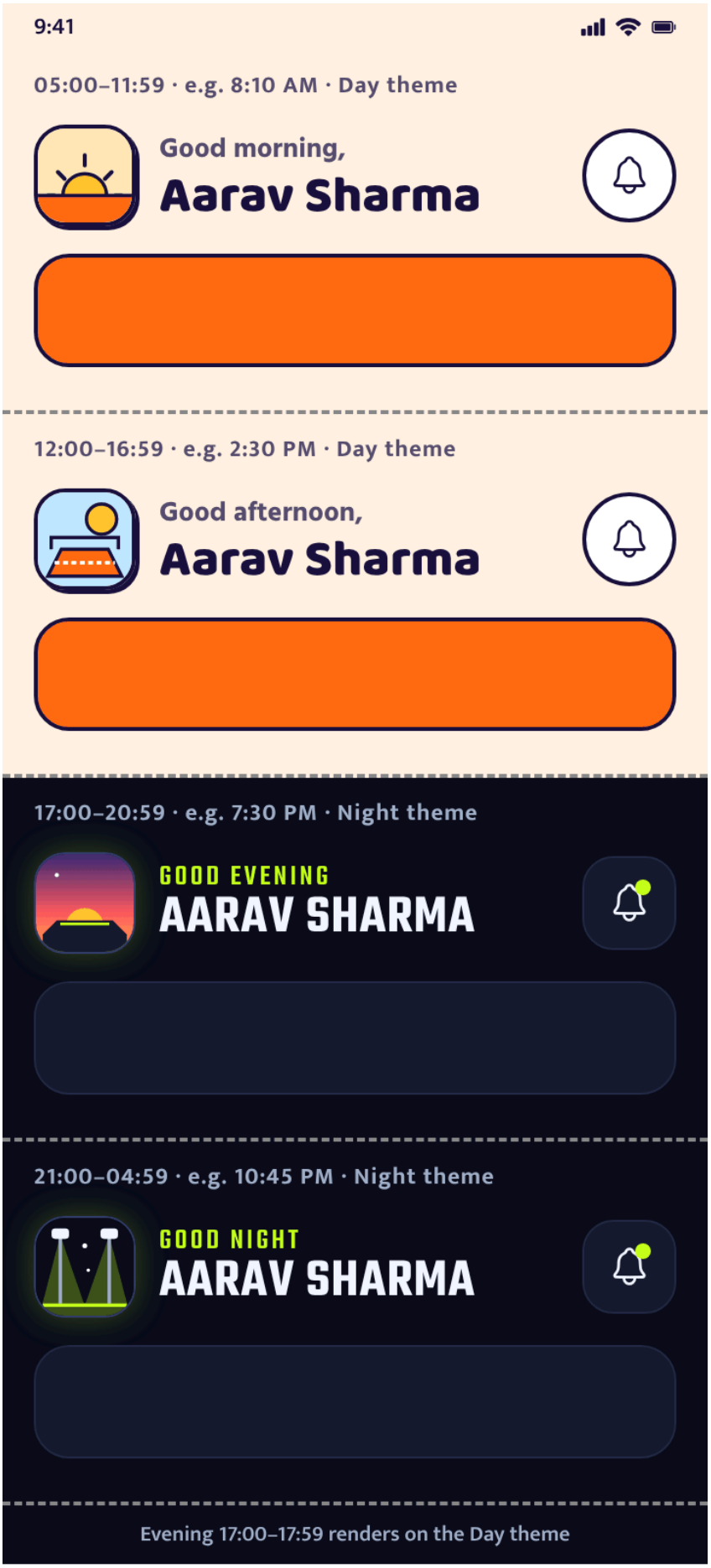
Component	Anatomy / size	States	Day vs Night
OTP cells	6 × 46 × 58	empty / focus / filled / error shake 240 ms	as input
Stepper	dots 10, active 28 wide, "Step n of N"	done / current / todo	–
List row	≥ 64, leading 40 – 44	rest / pressed / unread	dividers: Day dashed, Night solid
Tab bar	4 tabs + centre FAB	active / inactive	Day Ball pill; Night Volt label
Live badge	h22 – 24, dot 7 pulsing 1.2 s (static with Reduce Motion)	–	–
Player chip (scoring)	min 96 × 64: number, name, points, position badge	rest / pressed / serving / disabled	–
Team panel (scoring)	half width, min-h 214, "Tap · team +1" hint	rest / pressed / serving	Day sticker panel; Night side-colour border / glow
Radio option card	min-h 76, thumb 44 × 52, radio 26	selected / unselected	Day Ball fill; Night Volt border
Switch	52 × 32	on / off / disabled	–
Bottom sheet	radius 24 top, grabber, rows ≥ 64, Cancel 52	open / dragging	scrim: Day ink 55 %; Night black 72 %
Toast (Undo)	h56, 8 above the tab bar	4 s auto-dismiss / action	–
Empty / error state	art 120 – 180, title, body, primary + text link	offline / server / permission	art recoloured by tokens
Skeleton	`surface2` blocks	shimmer 1.2 s (static with Reduce Motion)	–
Greeting header	art 52, greeting 14, name 26 – 28, bell 48	4 variants	see 11.6
Reaction button (viewer)	48 circle, emoji 22	rest / pressed / sent	UI only in 1.5.7

Kit components live in `src/components/DayNightKit.tsx`, `DNKit.tsx` and `ui.tsx`.

11.6 Day / Night logic and the greeting



- **Auto is time-based only** in 1.5.7 (Akshit's choice). Spec v2 said the phone's dark mode should win; the code deliberately does not follow that.
- The appearance screen shows *"Now showing … – switches to Night at 6:00 PM"*.
- **Live screens hold the switch**: the scorer screen calls `holdTransitions(true)`; a switch that fires during a rally is stored as pending and applied after the next point as a **400 ms crossfade**. Taps are never blocked.
- The full animation plays **once per switch**.



Greeting	Hours	English text
Morning	05:00 – 11:59	"Good morning, {name}"

Greeting	Hours	English text
Afternoon	12:00 – 16:59	"Good afternoon, {name}"
Evening	17:00 – 20:59	"Good evening, {name}"
Night	21:00 – 04:59	"Late game, {name}?"

`{name}` is the first word of the display name; without a name the bare form is used ("Good morning", "Late game?"). The greeting refreshes at every minute boundary (`src/theme/resolve.ts`). The greeting hours are separate from the theme hours (e.g. 17:00 is "Good evening" in the Day theme).

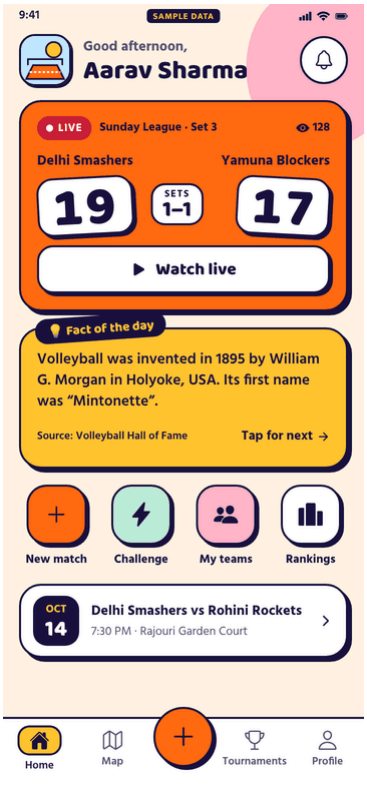
Sunset → Floodlight transition

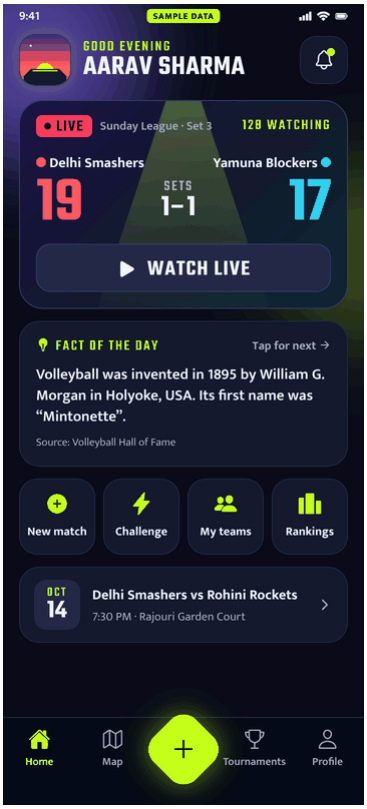


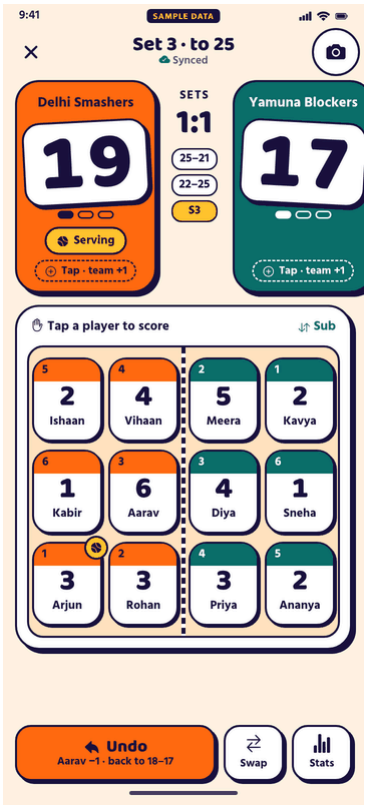
t (ms)	Day → Night
0 – 350	veil fades in over the Day UI; the sun starts to sink
350 – 750	sky goes day → dusk; the sun sinks behind the court; the overlay covers the screen by 700
750 – 1050	lamps flicker on; beams sweep; **at 850 ms the tokens are swapped underneath**
1050 – 1400	beams settle; the overlay fades out, revealing Night

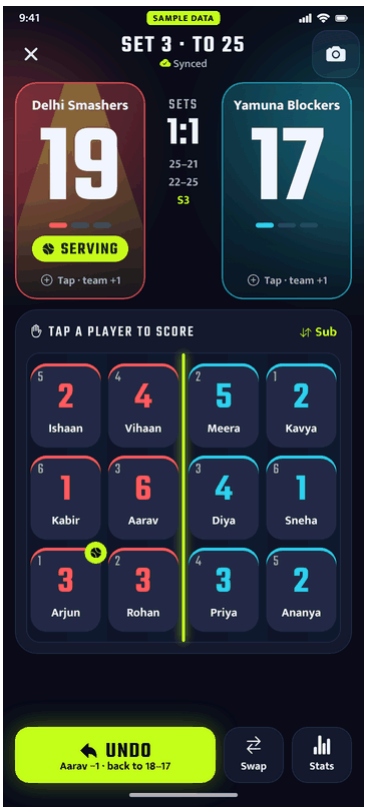
Sunrise plays the same timeline in reverse. Everything runs on the UI thread (reanimated); haptic *Light* when the lights come on; screen readers hear "Night theme on".

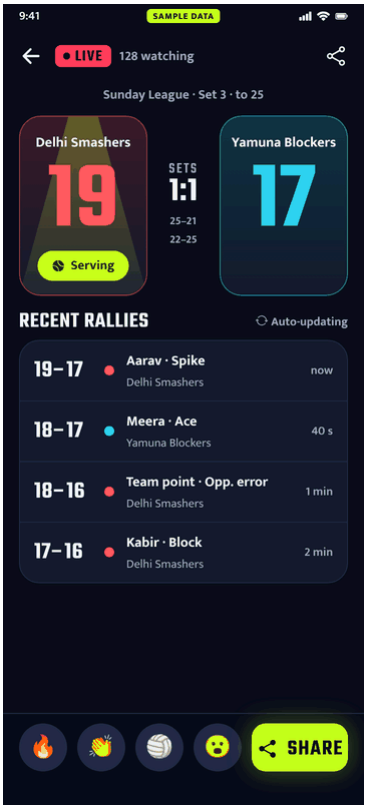
11.7 Screens in both themes

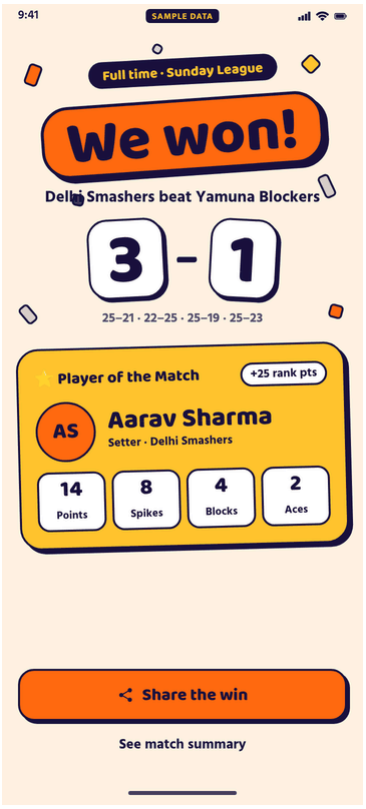

9:41
SAMPLE DATA
Good afternoon,
Aarav Sharma
LIVE
Sunday League · Set 3
128
Delhi Smashers
Yamuna Blockers
19
SETS 1–1
17
Watch live
Fact of the day
Volleyball was invented in 1895 by William G. Morgan in Holyoke, USA. Its first name was “Mintonette”.
Source: Volleyball Hall of Fame
Tap for next →
New match
Challenge
My teams
Rankings
OCT 14
Delhi Smashers vs Rohini Rockets
7:30 PM · Rajouri Garden Court
Home
Map
Tournaments
Profile

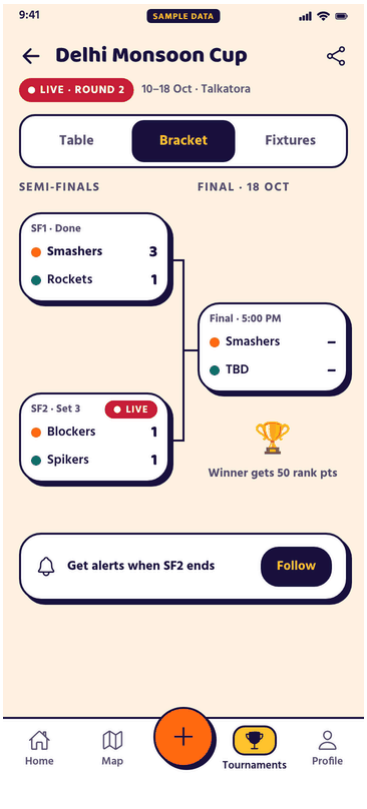

9:41
SAMPLE DATA
GOOD EVENING
AARAV SHARMA
LIVE
Sunday League · Set 3
128 WATCHING
Delhi Smashers
Yamuna Blockers
19
SETS 1–1
17
WATCH LIVE
FACT OF THE DAY
Tap for next →
Volleyball was invented in 1895 by William G. Morgan in Holyoke, USA. Its first name was “Mintonette”.
Source: Volleyball Hall of Fame
New match
Challenge
My teams
Rankings
OCT 14
Delhi Smashers vs Rohini Rockets
7:30 PM · Rajouri Garden Court
Home
Map
Tournaments
Profile

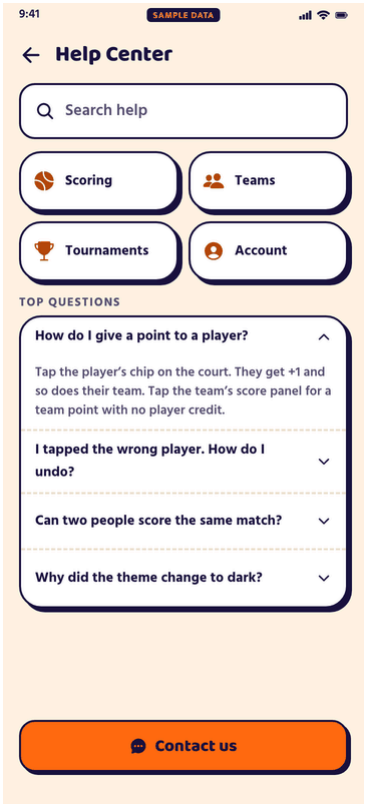

9:41
SAMPLE DATA
Set 3 · to 25
Synced
Delhi Smashers
19
Serving
Tap · team +1
SETS
1:1
25–21
22–25
S3
Yamuna Blockers
17
Tap · team +1
Tap a player to score
Sub
5 2 Ishaan
4 4 Vihaan
2 5 Meera
1 2 Kavya
6 1 Kabir
3 6 Aarav
3 4 Diya
6 1 Sneha
1 3 Arjun
2 3 Rohan
4 3 Priya
5 2 Ananya
Undo
Aarav −1 · back to 18–17
Swap
Stats


9:41
SAMPLE DATA
SET 3 · TO 25
Synced
Delhi Smashers
19
SERVING
Tap · team +1
SETS
1:1
25–21
22–25
S3
Yamuna Blockers
17
Tap · team +1
TAP A PLAYER TO SCORE
Sub
5 2 Ishaan
4 4 Vihaan
2 5 Meera
1 2 Kavya
6 1 Kabir
3 6 Aarav
3 4 Diya
6 1 Sneha
1 3 Arjun
2 3 Rohan
4 3 Priya
5 2 Ananya
UNDO
Aarav −1 · back to 18–17
Swap
Stats









App screenshots (1.4.24 – 1.4.26 builds, demo data):



11.8 Screen-by-screen design notes

General rules: scroll content gets bottom padding = tab bar (88) + 16 or docked button + 48; input screens keep the primary button **above the keyboard**; one primary action per screen; 48 dp targets; Hindi may wrap to 2 lines but is never truncated in buttons.

#	Screen	Primary action	Notes	In 1.5.7?
1	Onboarding (3 swipes)	Get started	skip, page dots, "Log in" link	not built
2	Login (+91 number)	Send OTP	field and button stay above the keyboard; language switch	yes
3	OTP	Verify (disabled until 6 digits)	6 cells, resend timer, change number	yes

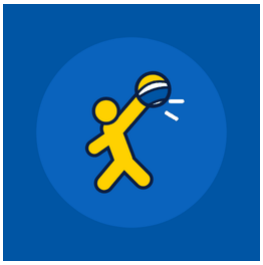
#	Screen	Primary action	Notes	In 1.5.7?
4	Profile setup	Let's play	photo (optional), name, role chips, city, DOB picker	yes
5	Home	Watch live (+ FAB New match)	greeting header, live hero, fact card, quick actions, upcoming	yes
6	Teams list	Create team	search, rows ≥ 64, captain tag	yes
7	Team detail	Challenge	stats, roster, add by mobile number	yes
8	Create team	Save team	name, colour (6 swatches × 48), players by number	yes
9	New match flow	Next	spec: 4 steps with shirt numbers; **app: 3 steps (Teams → Rules → Start)**	changed
10	Live scoring	tap player = player +1 & team +1; tap panel = team +1	sticky Undo, camera, serving, sets won	yes
11	Live viewer	Share	read-only scoreboard, recent rallies, reactions (UI only)	yes
12	Match summary / POTM / win	Share the win	Day confetti stickers / Night ray burst	yes
13	Photos sheet	Take photo	Take photo, Gallery, Cancel	yes
14	Tournaments list	New tournament	live / upcoming / finished cards	yes
15	Tournament detail	Follow	segmented Table / Bracket / Fixtures	yes
16	Rankings	–	spec: Players / Teams, period chips, podium, pinned own row	partly
17	Profile	Edit profile	rank card, stats grid, form; Settings → Appearance	yes
18	Appearance	(selection saves)	Auto / Day / Night radio cards + animation switch	yes
19	Help Center	Contact us	search, topic tiles, FAQ accordion	yes
20	Notifications	Mark all read	Today / Earlier, inline Accept / Decline	not built (push + Home cards instead)
21	Empty / loading / error	Create / – / Try again	empty hero, Home skeleton, offline / server / permission	partly

11.9 Logo history

v1 – ten concepts (13 Oct 2026)



The first four concepts used the palette **blue `#0055A4`, yellow `#FFD100`, ink `#0B1F44`**: 1 Tricolour Ball, 2 Live Pulse, 3 VL Monogram, 4 Spike (the first recommendation was the Tricolour Ball). Six more were added: 5 Shield, 6 Net, **7 Spiker**, 8 Play Ball, 9 Flame, 10 Rangoli.

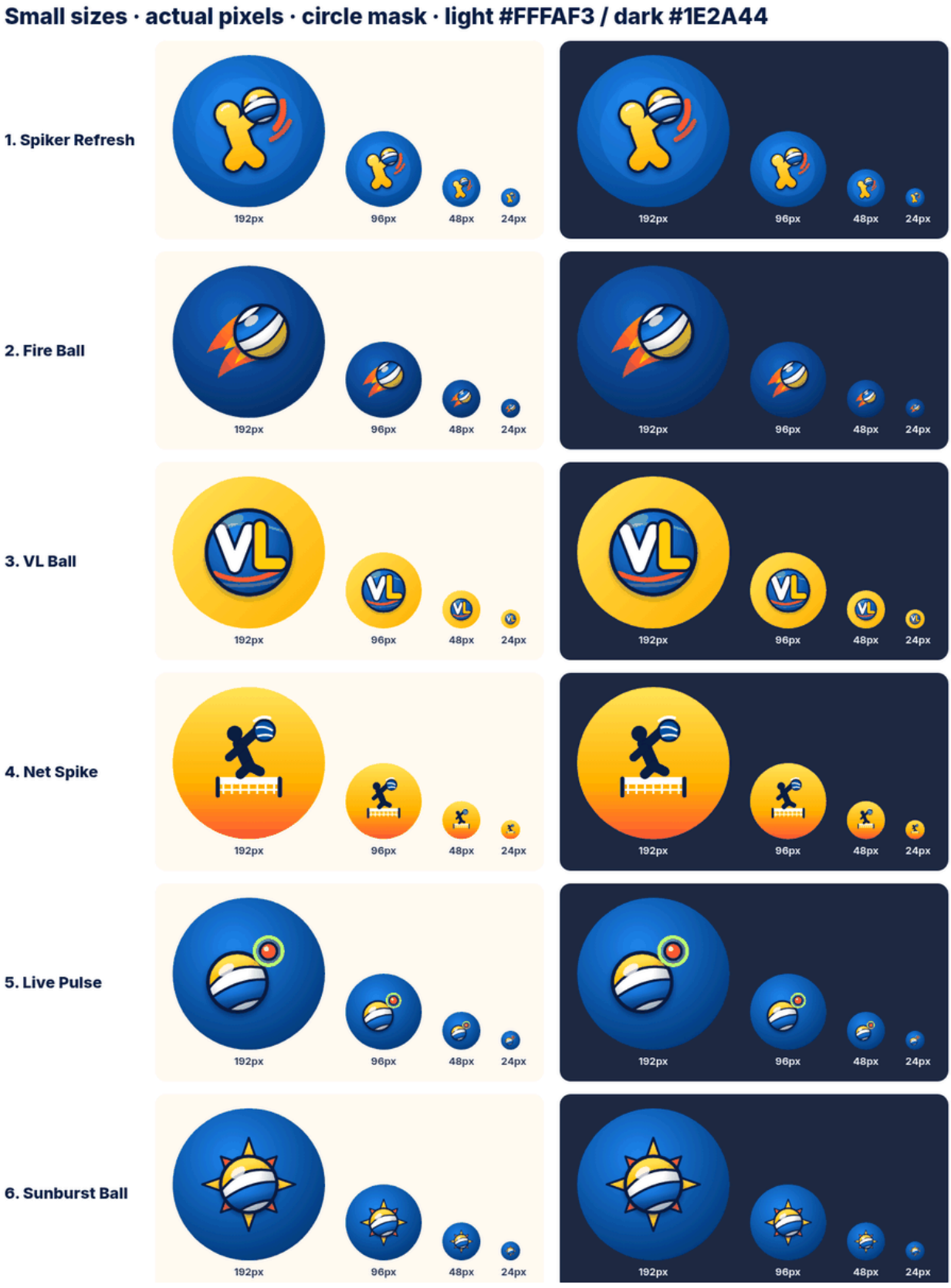


7 · Spiker was chosen ("final-spiker"). It is the icon that ships in 1.5.7 (`assets/icon.png`, adaptive background `#0055A4`, monochrome layer, Day / Night splash, white notification icon).

v2 – six refreshed options (29 Oct 2026)



1 Spiker Refresh, 2 Fire Ball, 3 VL Ball, 4 Net Spike, 5 Live Pulse, 6 Sunburst. At 24 px, VL Ball and Spiker read best. **Top pick: Spiker Refresh** – "the most ownable idea (a person playing, not just a ball)", the strongest colour mass at 48 px, and it keeps the Spiker Akshit liked.



Final – the Bold Spiker (chosen, for build 1.5.8)







Akshit then chose a bolder version: *a big glowing yellow athlete spiking a blue / yellow / white ball, with motion streaks, on a blue radial gradient* – original vector artwork based on the chosen concept image. The full kit is in `volleylive-design/logo/v2/final-bold-spiker/`. **It is not in 1.5.7**; adopting it is a file swap plus a native rebuild (`expo prebuild`) in 1.5.8.

Bold Spiker: adaptive masks (pink = 66dp safe zone)



108dp canvas · safe zone



Round



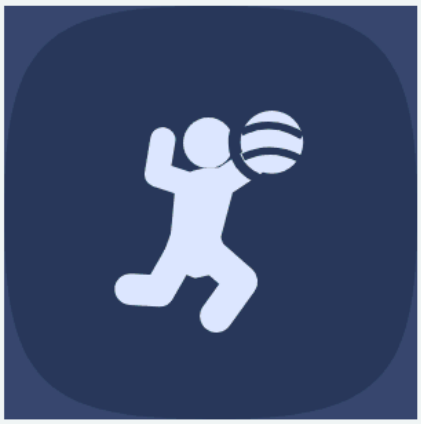
Squircle



Themed mono · safe zone



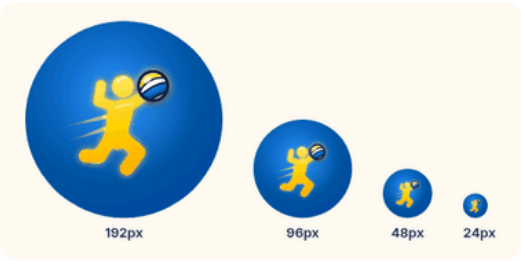
Themed · Round



Themed · Squircle

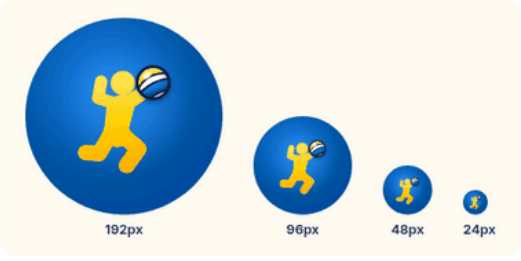
Bold Spiker: small sizes, actual pixels, circle mask, light #FFFAF3 / dark #1E2A44

Launcher art (adaptive)

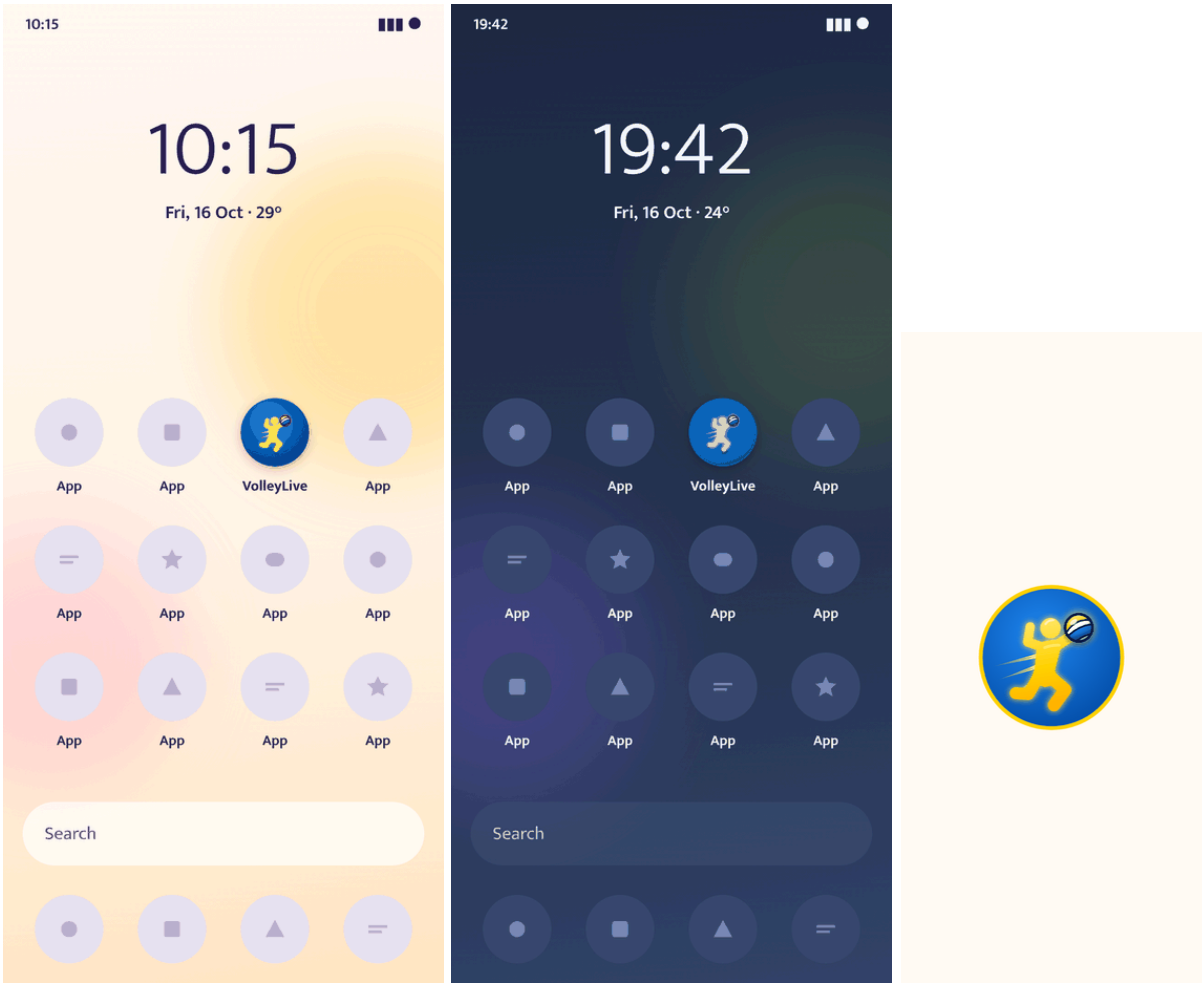


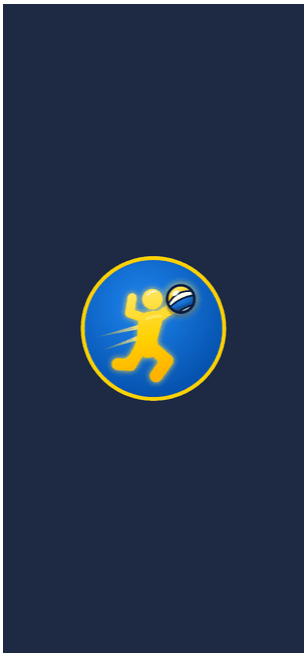


Small variant (no streaks/glow, thicker limbs): favicon, ≤24px uses









11.10 Icon specifications (Bold Spiker)

Geometry

- **Adaptive canvas 108 × 108 dp**; **safe zone: 66 dp circle → 21 dp padding per side (19.44 %)**.
- Key art (figure + ball, glow excluded) fits within a 33.1 dp radius; only faded streak tips and glow pass the safe zone (checked under round and squircle masks).

- Monochrome layer: same geometry. Notification icon: silhouette cropped to a 62 dp viewBox (~85 % of the 24 dp grid).
- Small variant (favicon): no streaks or glow, limbs 1.4 dp thicker, for 16 – 32 px.

Colours

Use	Hex
Brand blue (adaptive `backgroundColor`, notification `color`)	`#0055A4`
Background gradient highlight / mid / edge	`#1E86F0` / `#0055A4` / `#0047A0`
Background glow	`#3FA0FF` 45 % → 0 %
Figure gradient top / mid / bottom	`#FFE45A` / `#FFD100` / `#FFB300`
Figure glow	`#FFD54A`
Streaks	`#FFF3B0` → `#FFD100` → 0 % (Day logo mark `#0055A4` → 0 %)
Ball panels	yellow `#FFE45A` → `#FFC400`, blue `#1A7BE0` → `#003F86`, white `#FFFFFF`
Ink outline	`#0B1F44`
Badge ring	`#FFD100`
Splash background Day / Night	`#FFFAF3` / `#1E2A44`
Monochrome and notification	`#FFFFFF` on transparent (system-tinted)

Files and sizes

Asset	Files	Size / format
Adaptive foreground	`adaptive/foreground-1024.png`, `foreground-432.png`, `adaptive-foreground.svg`	1024 / 432 px, transparent
Adaptive background	`adaptive/background-1024.png`, `background-432.png`, `adaptive-background.svg`	gradient image (or the single colour `#0055A4`)
Legacy icon	`legacy/icon.png`	1024 px, RGB, no alpha
Play Store icon	`legacy/play-store-512.png`	**512 × 512**, 32-bit RGBA, fully opaque, full square, no rounding
Monochrome (Android 13 themed)	`monochrome/monochrome-1024.png`, `-432.png`, `.svg`	white on transparent
Splash	`splash/splash-icon.png` (+ Day / Night previews)	1024 px transparent badge; `imageWidth` 200
Notification	`notification/notification-icon-96.png`, `notification.svg`, `drawable-mdpi` 24 · `hdpi` 36 · `xhdpi` 48 · `xxhdpi` 72 · `xxxhdpi` 96 (`ic_notification.png`)	white on transparent
Favicon	`favicon/favicon.ico` (16 / 32 / 48), `favicon-16/32/48.png`	small variant
In-app logo	`logo/logo-badge`, `logo-mark-day`, `logo-mark-night` – `.svg`, `.png` 120, `@2x` 240, `@3x` 360	badge / marks

`app.config.ts` keys for 1.5.8

Key	Value
`android.adaptiveIcon.foregroundImage` / `backgroundImage` / `monochromeImage`	the adaptive / monochrome files above
`android.adaptiveIcon.backgroundColor`	`#0055A4`

Key	Value
`expo-splash-screen`	`image` = splash icon, `imageWidth` 200, `backgroundColor` `#FFFAF3`, dark `#1E2A44`
`expo-notifications`	`icon` = notification icon, `color` `#0055A4`
`icon`	`legacy/icon.png`
`web.favicon`	`favicon/favicon-48.png` copied to `assets/favicon.png` (optional `favicon.ico` in `public/`)

Integration steps are in `final-bold-spiker/INTEGRATION.md`.

12. Glossary

Term	Meaning
Abandon	ending a match without a result. With no rally it is immediate; after the first rally the interim policy **'review'** records it for review. See `docs/LIFECYCLE.md`.
Abandon policy	`review` (current, interim), `flag` or `consent` – set in `src/core/abandonPolicy.ts`.
Admin	a user id listed in `ADMIN_UIDS` on the server (or the demo admin in demo builds). Reviews flagged matches, manages daily items, feedback, support and Ask content.
Anti-fraud limits	rules that stop a result from counting: shorter than 10 minutes (flagged), more than 3 counted matches per player per IST day, more than 2 per team pair per IST week.
API backend	the app talking to the Node / Express / MySQL server.
Ask VolleyLive	the read-only in-app assistant (RAG + read-only tools).
Audit (Ask)	`ask_audit` rows: outcome, tools, HMAC hash of the question.
Bold Spiker	the chosen v2 app icon, planned for build 1.5.8.
Captain	the team's leader; edits the team, sends / answers challenges, confirms or rejects results.
Captain verdict	the opposing captain's confirm / reject of a finished match.
Card (Ask)	a structured result (match, standings, leaderboard, player, team, list) built only from tool data.
Challenge	a captain's invitation to another team to play at a day, time and place.
Counted	a result that is confirmed by the opposing captain and not flagged / voided; only counted results feed rankings and the wallet.
Credit	the player a team point is attributed to (with an optional shot type: spike, block, ace, tip).
Day / "Sunday Court"	the light theme (Court orange, Ball yellow, ink).
Defang	neutralising instruction-like text inside data before it reaches the model.
Demo backend / demo mode	data only on the phone; any number, code `123456`.
Deciding set	the last possible set (5th in Indoor, 3rd in Beach), usually to 15.
Dev auth	unsigned development tokens; local testing only.
DLT	India's SMS sender registration required for OTP messages.
DPDP	India's Digital Personal Data Protection Act.
EAS	Expo Application Services (cloud builds).
Epoch (reopen / verdict)	lifecycle counters that make an admin re-open invalidate older verdicts.
Fact / Quote of the day	daily items on Home; admin-managed; tombstoned instead of deleted.
FIVB	the international volleyball federation; VolleyLive's rule notes are its own words with a link to the FIVB rules.
Fixture	a scheduled tournament match between two tournament teams.
Flagged	a result held for admin review (e.g. too short).
Floodlight / Night	the dark theme (navy, Volt green, Ultra violet).
Groups + knockout	tournament format: round robin in groups A – H, then a knockout bracket.
Held-out (eval)	Ask eval questions frozen before the first run and never tuned against.
HMAC	keyed hash (SHA-256 with `ASK_AUDIT_HMAC_KEY`) used for the Ask audit.

Term	Meaning
IST	India Standard Time, UTC+05:30 – used for daily / weekly limits.
Knockout	tournament format: lose and you are out; byes to top seeds.
League	round robin; 3-0 / 3-1 = 3 pts, 3-2 = 2 pts, 2-3 loss = 1 pt.
LiveBus	the server's in-process publish / subscribe for live updates (Redis for scale-out).
Masking (PII)	replacing phone numbers and e-mails in Ask output and stored text.
Mock provider	the default, deterministic Ask answer step with no network or key.
OTP	one-time password: the 6-digit sign-in code.
Participant	anyone attached to a match (players, creator, scorer).
POTM	Player of the Match – auto-picked by most credited points, confirmable by the scorer; +25 ranking points.
Progress (save)	the scorer's rally save (`PUT /matches/:id/progress`) with a client sequence number.
RAG	retrieval-augmented generation: answer from retrieved VolleyLive content with sources.
Rally	one point played; each rally adds one event to the match log.
Ranking points	credited points + 10 per win + 25 per POTM, over finished counted matches; 3 matches needed to be ranked.
Registered player / team	a VolleyLive user / a saved team with accepted members.
Rotation	the order of the six positions; advances on side-out.
Rule set	Indoor, Beach or Custom match rules.
Scorer	the only person allowed to score a match.
Scorer hand-off	passing the scorer role to another eligible person.
Seed	a team's ranking position for draws and byes.
Side-out	the receiving team wins the rally and gains the serve.
Spiker	the v1 app icon (concept 7) that ships in 1.5.7.
Terms gate	every write requires the current Terms version to be accepted.
Tombstone	a deleted daily item kept as a blank, deleted row.
Tool (Ask)	one of 10 read-only data functions the assistant may call.
Undo	removes the last rally while the match is live.
Viewer	the read-only live scoreboard for everyone except the scorer.
Viewer-scoped	running with the signed-in user's own permissions.
Volt	Night primary green `#C6FF1A`.
Wallet	in-app points (no money value): match points + bonus (+100 every 5 counted wins).

13. FAQ

Players

How do I sign in? Enter your 10-digit mobile number, tap **Send OTP**, type the 6-digit code. First time: accept the Terms and set up your profile.

I didn't get the code. Wait for the resend timer and tap **Resend code**. Check the number with **Change**. There is a per-hour limit on codes per number.

Can other players see my phone number? No. They can add you to a team by number, but your number is never shown to them, and Ask never reveals it.

How do I give a point to a player? On the scoring screen tap the player's tile: the player and their team both get the point. Tap the team's half for a team point only (e.g. an opponent error).

I tapped the wrong player. Tap **UNDO** – it removes the last rally. You can also correct the credit on one of the last 3 rallies after the match.

Why can't I undo after the match ended? Once the winning rally is in, the match is final everywhere.

Can two people score the same match? No. Only the assigned scorer can score; the scorer can hand over the role. Everyone else follows live.

Why doesn't my win count yet? A result counts only after the **opposing captain confirms** it, and only if it is not flagged (shorter than 10 minutes) and within the daily / weekly limits. The wallet shows what is waiting.

How are ranking points calculated? Credited points + 10 per win + 25 per Player of the Match, over finished counted matches. You need 3 finished matches to appear in the rankings.

Are wallet points money? No. They are in-app points only, with no money value.

Why did the theme change? In **Auto**, the app uses Day from 06:00 to 17:59 and Night from 18:00 to 05:59. Choose Day or Night in Settings → Appearance to fix it. The animation can be turned off there.

Does the app work offline while scoring? Yes. Rallies are saved on the phone ("Offline – saved on phone") and sent in order when the network is back.

How do I switch to Hindi? Settings → Language → हिन्दी (also on the login screen).

How do I delete my account? 1.5.7 has no in-app delete button yet. Contact support through Help → Contact us. The server supports owner-only deletion (`DELETE /api/users/:id`).

Captains and organisers

How do I challenge another team? Home → Challenge (captains only), pick your team, find the other team, set day, time, place and an optional message.

Who scores a challenge match? The accepting captain is the scorer; the challenger's captain confirms the result.

How many teams can a tournament have? Up to 32 registered teams with at least 6 players each. Groups + knockout needs at least 4 teams.

How are league points awarded? 3-0 / 3-1 win = 3 points, 3-2 win = 2 points, 2-3 loss = 1 point.

Can I delete a match? Only a match that never had a rally (creator or admin). A played match cannot be deleted.

Ask VolleyLive

What can Ask do? Answer rules and how-to questions with sources, and show your stats, teams, matches, live scores, tournament tables and fixtures, challenges and the leaderboard.

Can Ask change my data? No. It is read-only; it tells you which screen to use instead.

Is Ask a real AI model? By default it runs a deterministic mock planner. An operator can connect OpenAI, Gemini or Anthropic on the server with a key in the server's secret store; this has not been switched on yet.

Who can see my Ask history? Only you. **Clear** deletes it.

Developers and operators

Which backend does a build use? Decided from `EXPO_PUBLIC_*` variables: API if `EXPO_PUBLIC_API_URL` is set, Firebase if Firebase keys are set, otherwise demo. See 03.

Where do secrets go? Never in the repo. Use the EAS environment, the host's secret store or a git-ignored `.env`. `.env.example` files are placeholders.

Can I edit an applied migration? No – the runner checks sha256 checksums and refuses to start. Add a new migration.

Why did `adb install` fail with `INSTALL_FAILED_VERSION_DOWNGRADE`? The installed app has a higher versionCode. Build with a higher versionCode; do not uninstall (that wipes app data).

Why `INSTALL_FAILED_VERIFICATION_FAILURE` on a Realme phone? Enable Developer options → *Install via USB* and approve the prompt on the phone.

How do I run more than one API instance? Set `REDIS_URL` (shared rate limits and live bus) and use S3 or a shared volume for photos.

14. Feature-by-feature acceptance checklist

Use this list to accept a build (manual walk on a phone or the web build at 360 px, in **Day and Night** and **EN and HI**). Each line is a pass / fail check. Automated coverage is noted where a suite proves it.

A. Sign-in and profile

- [ ] A1 Login accepts a valid 10-digit number and rejects an invalid one with a clear message.
- [ ] A2 **Verify** stays disabled until 6 digits are entered; a wrong code shows an error; resend unlocks after the timer.
- [ ] A3 First sign-in shows the Terms; continuing requires ticking the box.
- [ ] A4 Profile setup requires a name; DOB below `MIN_AGE` is refused.
- [ ] A5 Demo builds show the demo banner; production builds never do (`check:strip`).
- [ ] A6 Sign out returns to Login; the session no longer works on the server.
- [ ] A7 Profile shows stats, recent form, teams, history, memories; private details only to the owner.

B. Settings

- [ ] B1 Appearance: Auto / Day / Night apply immediately; Auto follows 06:00 / 18:00.
- [ ] B2 The Sunset → Floodlight animation plays once per switch; with the switch off or Reduce Motion it fades.
- [ ] B3 On the scoring screen a theme switch never covers the score (crossfade after the rally).
- [ ] B4 Language EN ↔ HI switches every screen; no truncated Hindi in buttons at 360 px.
- [ ] B5 Wallet shows balance, bonus rule, next bonus, pending / flagged counts and history.

C. Teams

- [ ] C1 Create a team with name and colour; creator is captain.
- [ ] C2 Add players by number; an invited player can accept or decline.
- [ ] C3 Phone numbers of other players are never displayed.
- [ ] C4 A team needs 6 players to play; at most 12.
- [ ] C5 Unconfirmed results are not counted on the team page ("waiting for the other captain").

D. New match and live scoring

- [ ] D1 3-step flow Teams → Rules → Start; **Next** explains what is missing.
- [ ] D2 Exact 6 forces picking 6 for larger teams (captain always plays); Custom allows 6 – 12.
- [ ] D3 A player cannot be in both teams.
- [ ] D4 Indoor and Beach presets match the documented rules; editing makes the rules Custom.
- [ ] D5 Tap a player = player + team point; tap a team half = team point only.
- [ ] D6 Sets end at the right score (win by 2); set point / match point labels; deciding set to 15.
- [ ] D7 Side switches prompted at the rule's points; timeouts limited per set; substitutions work.
- [ ] D8 Serve indicator and rotation advance correctly.

- [ ] D9 Offline: rallies are kept and sync in order; status shows Offline / Syncing / Synced.
- [ ] D10 Only the scorer can score; others see "Watch live".
- [ ] D11 The winning rally ends the match; the result screen offers Share and Summary.
- [ ] D12 Viewers see the score update after every rally without refreshing.
- *Automated: scoring, rotation, rules, replay and lifecycle suites (Jest, server, rules).*

E. Undo

- [ ] E1 Undo removes the last rally (team point and player credit) and shows the new score.
- [ ] E2 "Nothing to undo yet" at 0 rallies.
- [ ] E3 Undo is refused once the match is over and after POTM is set.
- [ ] E4 Undo back to 0 does not make a played match deletable.

F. Results, POTM and anti-fraud

- [ ] F1 POTM auto-picks the most credited points; the scorer can change and confirm it.
- [ ] F2 Only the opposing captain can confirm or reject; reject reason ≤ 200 characters.
- [ ] F3 A match under 10 minutes is flagged and does not count until reviewed.
- [ ] F4 Daily (3 per player) and weekly (2 per team pair) limits apply in IST.
- [ ] F5 Admin approve only clears the flag; void never counts; admins cannot review their own matches.
- [ ] F6 Rankings and wallet count confirmed results only.

G. Abandon

- [ ] G1 A match with no rally can be abandoned or deleted.
- [ ] G2 After the first rally the 'review' policy records the abandon for review; the opposing captain can report.
- [ ] G3 A won match cannot be abandoned.

H. Challenges

- [ ] H1 Only captains can send; a team cannot challenge itself; both teams need 6 players.
- [ ] H2 Challenged captain gets a notification and can accept / decline.
- [ ] H3 Accept creates a playable match (accepting captain scores; challenger's captain confirms).
- [ ] H4 A challenge can only be linked to a match between its two teams.

I. Tournaments

- [ ] I1 Knockout: byes to top seeds; play-in round when needed; bracket readable at 360 px.
- [ ] I2 League: table uses 3 / 2 / 1 points.
- [ ] I3 Groups + knockout: ≥ 4 teams, ≤ 32; seeded and random draws; preview counts match.
- [ ] I4 Only the organiser (or an admin) changes a tournament.
- [ ] I5 Table / Bracket / Fixtures tabs update from match results; champion shown.

J. Home and map

- [ ] J1 Greeting matches the hour (morning / afternoon / evening / night) with the first name.
- [ ] J2 Fact / Quote of the day shows a source; "Tap for next" cycles; Share works.
- [ ] J3 Quick actions open the right screens (Courts → Map tab).
- [ ] J4 Live hero, rank card, invites, challenges and upcoming are shown when relevant.

- [ ] J5 Map shows courts and games; distance chips filter; Play nearby works without location permission.

K. Ask VolleyLive

- [ ] K1 Rules questions answer with sources (EN, HI, romanised HI).
- [ ] K2 "My stats / teams / matches", live score, standings, fixtures, challenges, leaderboard show correct cards.
- [ ] K3 Write requests are refused with a pointer to the right screen.
- [ ] K4 Requests for someone else's phone / private data are refused.
- [ ] K5 Prompt-injection attempts are refused; injected text in data is ignored.
- [ ] K6 Phone numbers and e-mails never appear in answers or stored history.
- [ ] K7 History is owner-only; Clear deletes it.
- [ ] K8 Over the rate limit the user sees a friendly "try again later".
- *Automated: Ask eval (≥ 98 % at 1.5.7), server Ask tests, PII tests.*

L. Help, feedback, rate, about

- [ ] L1 Help Center search, topics and FAQ work in EN and HI; Contact us creates a ticket.
- [ ] L2 Support buttons only appear for configured contacts.
- [ ] L3 Feedback sends 1 – 5 stars + optional text; rate-limited; no rewards.
- [ ] L4 Rate opens the Play review sheet in the Play build; the pre-prompt is off by default.
- [ ] L5 About shows version, versionCode and backend; Licences and Privacy (DRAFT) open.

M. Non-functional

- [ ] M1 Contrast: every text pair passes WCAG AA in Day and Night (contrast scan 0 failures).
- [ ] M2 Touch targets ≥ 48 dp; screen reader labels on tiles, chips and buttons.
- [ ] M3 Layout at 360 px: nothing clipped; content clears the tab bar and keyboard.
- [ ] M4 All suites green: Jest (+ 5 time zones), server, rules.
- [ ] M5 No secrets in the repo or the APK; production build passes `assertProductionBuild`.
- [ ] M6 The APK installs over the previous build with `adb install -r` (versionCode increased, same certificate).